

Finite Transfers and Open Energy Balances

Physical tests of loaded gears, switched returns, and prepared states

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Abstract

A rotating planet brought to a fourth central shaft gives a specific apparatus for measuring how an attached receiver changes the other shaft works. The ideal single-receiver control changes carrier work by $7/3$ joule in one second. We derive its local forces and identify the additional joint, support, stator and endpoint observations needed for a physical comparison. A dimensioned Oldham coupling resolves a different support model without assigning its forces to universal joints. An isolated-primary tapped-secondary circuit then makes switched return a definite connection problem, with persistent winding, cell, gate and thermal states. Exact controls distinguish clocked attraction from hybrid operation, terminal equivalence from hidden energy, and prepared energy from receiver work. Two series-cell preparations have the same entire terminal history but stores of 1 and 5 joules. Finite reconnection specifies binary observation and paired-work certificates; an LC/shunt example proves every charge-sign event on its full window. Reset order, material states and physical supplies lead to separate measurable questions. Independent uncertainties accompany work and endpoint observations. The results are exact or explicitly conditional within declared models; no apparatus data are used.

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Latest on GitHub: <https://github.com/hobnilre/physics-gear-op>

1 One apparatus, then its physical tests

A planet rolls between a sun and ring while its centre travels with the carrier. Bring the planet's rotating stub through two correctly phased universal joints to a separately supported central shaft. There are now four accessible shafts: sun, carrier, ring and planet output. When a receiver loads that fourth shaft, which reactions and work readings change, and how can their complete physical balance be measured?

Frames, Returns, and Port Power (Nilre and Herlin, 2026) constructs the ideal four-shaft mechanism and two electrical realizations of its terminal relations. Here the same construction supplies the first experiment. A held-ring control predicts a $7/3, \text{mathrm J}$ change in carrier work when one output receives $7/3, \text{mathrm J}$ in one second. Actual joint forces, moving supports, bearing losses and preparation must be identified independently. A separate Oldham model makes that distinction tangible: its known slot forces cannot determine a Cardan cross's reactions.

The electrical investigation likewise begins with connections that can be drawn and measured. A switched full-to-tap return is a different circuit from a change of voltage zero. A series-to-parallel bank change can reveal energy that its earlier terminal history concealed. Following these operations requires individual cells, finite winding states, switch and probe loading, gate supplies and actual reset endpoints.

The article proceeds from the four-shaft test to its mechanical qualifications, then to windings, switched returns and signed cell states. Later sections develop material and supply laws and the independent uncertainties needed for a decision. Exact controls establish their own predictions; unknown

constitutive laws require acquisition; a physical energy remainder requires simultaneous measurements of all included ports and endpoints. None of these tasks can supply the missing data for another.

2 Boundaries, signs, and exact integration

Power is positive into the named system. Mechanical products are $\mathbf{F} \cdot \mathbf{v} + \mathbf{M} \cdot \boldsymbol{\omega}$ at the force application point and body; electrical products are complete voltage differences times oriented current. Thermal export is $-T_b \dot{S}_{\text{out}}$ at boundary temperature T_b ; internal entropy production is not a second boundary heat port. On each explicit interval,

$$W_j(t_0, t_1) = \int_{t_0}^{t_1} e_j f_j dt, \quad r_E = E(t_1) - E(t_0) - \sum_j W_j. \quad (1)$$

For vector ports use the corresponding scalar product. Opposite sides of an internal interface are integrated separately and cancel only when their actual motions and efforts agree. An impulse represented by a declared jump law replaces a resolved finite edge; it is never added to that edge a second time. Both endpoints are constitutive evaluations, not residuals.

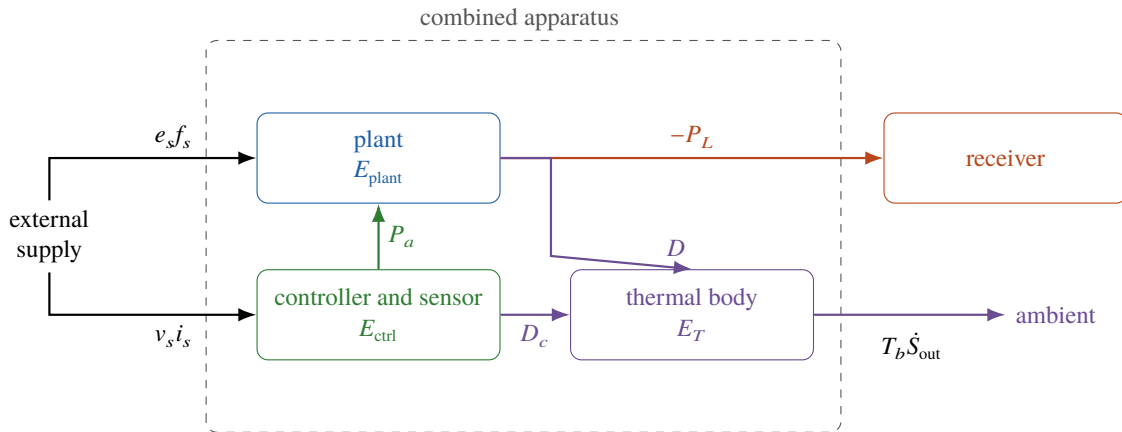


Figure 1. Schematic boundary hierarchy. Shared mechanical or electrical interfaces cancel only for the same realized path. Plant and controller conversion rates are D and D_c . Controller and thermal stores remain inside the combined apparatus; external supply, receiver, and ambient transfers remain measurable.

Figure 1 shows why enlarging a boundary changes the list of external transfers. All dimensional coefficients below are illustrative. In displayed worked paths, t denotes the numerical time in seconds and every coefficient has the SI units needed for the stated quantity. Unless another interval is specified, each worked path uses $[0, 1]$ s. Equations using symbolic T retain its time units. Configuration I is the simple train; II the linkage; III the compound geometries; IV the regular winding pair; V the prepared passive and switched circuits; VI the material and supply controls.

For later linear circuits a useful closed form integrates each port without solving for a missing work. Augment a constant, polynomial, or sinusoidal source by its exact differential equations, so that $\dot{y} = Ay$. For $P_j = y^T Q_j y$, with Q_j symmetric, define

$$\mathcal{I}(Q_j, A, y_0, T) = \text{vec}(Q_j)^T T \varphi_1(T(A \otimes I + I \otimes A)) \text{vec}(y_0 y_0^T), \quad \varphi_1(Z) = \sum_{n=0}^{\infty} \frac{Z^n}{(n+1)!}. \quad (2)$$

Here $y(T) = e^{AT}y_0$ and $Q_j = (e_j f_j^T + f_j e_j^T)/2$ when the two linear channel vectors are e_j, f_j . This entire matrix function is a closed form, including singular A ; no series is truncated.

Lemma 1 (Independent port primitive). *Equation (2) equals the signed integral of that port.*

Proof. For $Y = yy^T$, $\dot{Y} = AY + YA^T$. Column vectorization gives $\text{vec } Y(t) = e^{t(A \otimes I + I \otimes A)} \text{vec } Y(0)$. Integrating its convergent exponential series and contracting with Q_j gives the result without using any store. $\square$

The numerical integration residual throughout this article is exactly zero: no numerical integration is used. The model residual comprises omitted physical laws and ports. It remains unquantified unless a separate, explicit constitutive control calculates a particular contribution.

3 Gears: free bodies, stores, and physical lead-outs

3.1 The fourth shaft and its first measurable comparison

The planet body carries a rotating stub inside a carrier-mounted bearing. Two universal joints connect that stub to a separately supported central shaft o , beside the sun, carrier and ring shafts. The parallel-offset geometry has equal joint angles, with phasing fixed in the carrier's rotating joint plane. The intermediate shaft fixes that relative phase; equal angles and correct phasing are the ordinary double-joint conditions [Belden](#). With continuous joint lift $H_\beta(x)$ satisfying $\tan H_\beta(x) = \tan x / \cos \beta$, the map $\theta_o = \phi + H_\beta(H_\beta(\theta_p - \phi) + \pi/2) - \pi/2$ reduces to $\theta_o = \theta_p$. The later phase and support controls change specified parts of this model.

Four separate shafts; one rotating planet stub

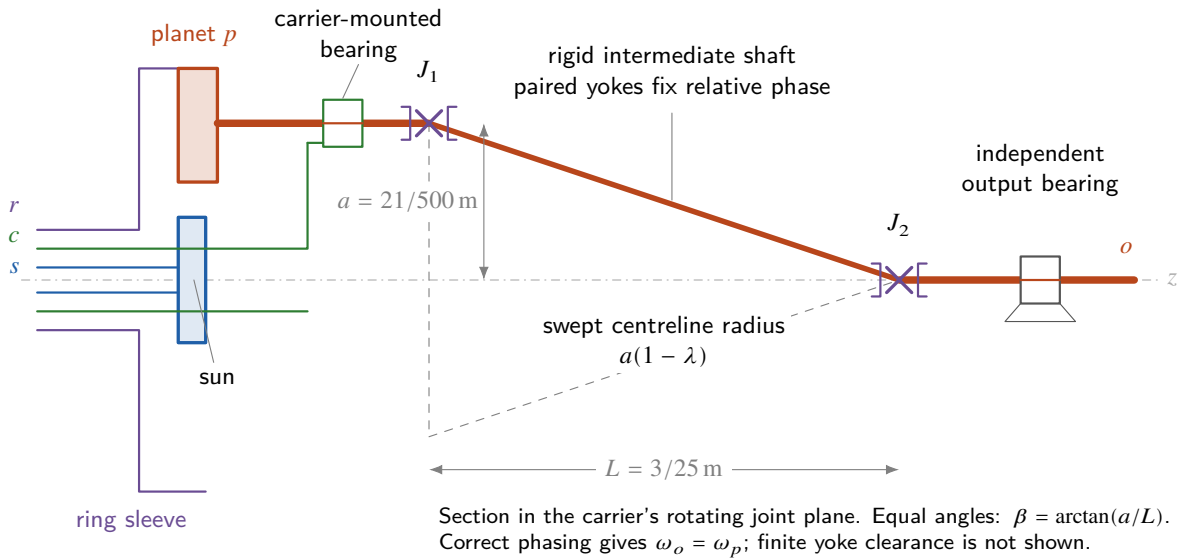


Figure 2. The recurring four-shaft apparatus. The output is connected to the rotating planet stub, not its carrier-fixed support pin. The yokes are phased relative to the rotating joint plane. Dimensions describe centreline geometry; complete joint forces, packaging and physical clearances require independent identification.

For teeth (24, 18, 60), module 1/500 m, offset 21/500 m and joint-centre axial separation 3/25 m, the

two mesh equations give

$$(\omega_s, \omega_c, \omega_r, \omega_o) = (u + d, u, u - 2d/5, u - 4d/3). \quad (3)$$

All torques below enter the assembly. For a single steady loaded planet, the sun force is $F_s = \tau_s/r_s$, planet spin balance gives $F_r = F_s - \tau_o/r_p$, and the tangential pin force is $-F_s - F_r$. The ring and carrier moment arms then give

$$\tau_r = \frac{5}{2}\tau_s - \frac{10}{3}\tau_o, \quad \tau_c = -\frac{7}{2}\tau_s + \frac{7}{3}\tau_o. \quad (4)$$

These load reactions were obtained before evaluating power. Hold the ring, maintain $(\omega_s, \omega_c) = (7/2, 1)$ rad/s and compare the unloaded planet with $\tau_o = 1$ N m, at measured $\tau_s = 2$ N m.

Table 1. The same established motion and sun effort with one attached receiver. Independent control authority and actual effort measurements are requirements of this comparison.

One-second observation	Unloaded output	Loaded output
Output rate, rad/s	$-7/3$	$-7/3$
Carrier torque, N m	-7	$-14/3$
Ring holding torque, N m	5	$5/3$
Sun work into assembly, J	7	7
Carrier work into assembly, J	-7	$-14/3$
Ring work into assembly, J	0	0
Output work into assembly, J	0	$-7/3$

Four effective unit inertias give $673/72$ J at each endpoint, with orbit counted once in the carrier coefficient. The four separate work integrals sum to zero. This initialized steady interval has no impulse and does not assign a preparation work. The later 7 J comparison uses three receivers or total output torque 3 N m; it is a different load. Equal planet speeds do not determine force sharing. Under equal bilateral contact stiffness, loads $(3, 0, 0)$ and $(1, 1, 1)$ N m have the same aggregate reactions but sun-force vectors $(250/3, 0, 0)$ and $(250/9, 250/9, 250/9)$ N. Individual contacts and pin couples need observations when they enter the boundary.

The first comparison uses synchronized shaft torque and unwrapped angle; the held-ring transducer checks its reaction although its work is zero. As an illustrative calibration requirement for the two carrier observations, take $|\hat{\tau}| \leq 8$ N m, $|\hat{\omega}| \leq 101/100$ rad/s, $u_\tau \leq 1/200$ N m and $u_\omega \leq 1/1000$ rad/s. The product bound derived below gives $2611/100000$ J for their difference. A total $1/10$ J difference budget leaves $7389/100000$ J for independently bounded timing, setting and other observation errors. This resolves the predicted $7/3$ J change if those requirements are met. A complete residual also needs every support, drive, thermal and endpoint contribution.

3.2 A dimensioned support model with a different coupling

The U-joint angle map leaves its spatial forces undetermined. A mirrored Oldham pair supplies a separate determinate parallel-axis control. Take offset $a = 21/500$ m, combined middle mass $m = 1/5$ kg, middle inertia $I_m = 1/500$ and output inertia $I_o = 1/250$ kg m². Orthogonal slots have half-stroke $3/50$ m, pad arm $b = 1/50$ m, bearing separation $\ell = 3/100$ m and overhang $h = 1/100$ m. Mirroring puts net pad forces in one plane; locating bearings hold axes parallel. Angular misalignment and axial force are

excluded. Bilateral preload is declared as 100 N; contact strength, preload compliance and friction require material data. The declared domain is $|\omega_c| \leq 1$ and $|\omega_p| \leq 7/3$ rad/s; fixed output bearings do zero ideal work.

Put $u = (\cos \theta, \sin \theta)$, $v = (-\sin \theta, \cos \theta)$, $P = a(\cos \phi, \sin \phi)$ and middle centre $M = (P \cdot v)v$, where θ is planet orientation and $\delta = \phi - \theta$. For constant shaft rates, differentiation and the translational and angular free bodies give

$$K_L = \frac{ma^2}{2} [(\omega_c - \omega_p)^2 \cos^2 \delta + \omega_p^2 \sin^2 \delta] + \frac{I_m + I_o}{2} \omega_p^2, \quad (5)$$

$$F_i = -ma[(\omega_c - \omega_p)^2 + \omega_p^2] \sin \delta v, \quad F_o = -2ma(\omega_c - \omega_p)\omega_p \cos \delta u, \quad (6)$$

$$Q_p = ma^2 \omega_c^2 \cos \delta \sin \delta, \quad Q_c = -ma^2[(\omega_c - \omega_p)^2 + \omega_p^2] \cos \delta \sin \delta. \quad (7)$$

The force sum is $m\ddot{M}$. Input couple $T_i = Q_p - \tau_{\text{rec}}$ follows from the angular equation. Near and far bearing forces are $-(1 + h/\ell)F_i$ and hF_i/ℓ ; their sum and bending moment close independently. The overhang moment is $(-hF_{i,y}, hF_{i,x}, 0)$. The two pad resultants are $F_n/2 \pm T/(2b)$ for the local resultant F_n and transmitted torque T . This bilateral idealization must be checked against physical preload and contact capacity.

At $(\omega_s, \omega_c, \omega_r, \omega_p) = (7/2, 1, 0, -7/3)$, $\tau_s = 2$, $\tau_{\text{rec}} = 1$ in SI units and initial $\theta = \phi = 0$, set $\tau_o = -T_i$. The train then requires $\tau_r = 5 - 10\tau_o/3$, $\tau_c = -7 + 7\tau_o/3$; its carrier drive additionally supplies Q_c . The whole boundary includes train and linkage, so the two sides of the planet and moving-pin transfers cancel only after their separate integrals.

Table 2. Each work is the integral of its specified shaft product. The second interval returns the linkage store; the first does not.

Interval, s	W_s , J	W_c , J	W_r , J	W_{rec} , J	Independent ΔK , J
$[0, 3\pi/40]$	$21\pi/40$	$-7\pi/20 - 2499/5000000$	0	$-7\pi/40$	$-2499/5000000$
$[0, 3\pi/5]$	$21\pi/5$	$-14\pi/5$	0	$-7\pi/5$	0

The train alone has steady 673/72 J under its declared effective unit inertias; this control treats planet spin as that train coefficient and adds the new middle/output terms in K_L once. Evaluation at the two ends, rather than an assumed periodic energy, gives the last column. The exact residual is zero and no event occurs. These slot and bearing forces do not determine Cardan-cross forces.

Physical stator mounting also changes a brake law. For a massless viscous receiver with $D = 1/2$ N m s, shaft rate $\omega_p = -7/3$ and stator rate ω_f , inward powers to the receiver are $D(\omega_p - \omega_f)\omega_p$ and $-D(\omega_p - \omega_f)\omega_f$, with exported heat $-D(\omega_p - \omega_f)^2$. On $[0, 1]$ s the three works are $(49/18, 0, -49/18)$ J for $\omega_f = 0$ and $(35/9, 5/3, -50/9)$ J for $\omega_f = 1$. A real reference drive must supply the second mounting.

The zero-cycle support result also depends on the load history. In the massless fixed-phase joint control take $\cos \beta_1 = 4/5$, $\cos \beta_2 = 3/5$, $\omega_p = 2$, $\omega_c = 1$ and $\psi = t$ on $[0, \pi]$ s. For $k = 3/4$, $\chi = k/(\cos^2 \psi + k^2 \sin^2 \psi)$. A passive variable torque demand $\mathcal{T} = 1/\chi$ gives $P_p = 2$, $P_c = 1/\chi - 1$ and $P_{\text{rec}} = -(1 + 1/\chi)$. Since $\int_0^\pi \chi^{-1} dt = 25\pi/24$, the works are $(2\pi, \pi/24, -49\pi/24)$ J. The empty store at both endpoints gives zero residual. The nonzero support work is derived for this load; the constant-load cycle null retains its separate scope. Unequal joint angles require a different axis/support arrangement from the equal-angle parallel-offset baseline.

3.3 Locking and release with actual endpoint states

A coaxial sphere inside an annular carrier isolates the locking question without imposing an orbital rolling constraint. Let $I_b = 2/5$, $I_c = 3/5 \text{ kg m}^2$ and viscous clutch $D = 1 \text{ N m s}$. Holding $\omega_c = 1 \text{ rad/s}$ with the ball initially at rest gives $\omega_b = 1 - e^{-5t/2}$. On $[0, 1] \text{ s}$, independently integrating drive $D(1 - \omega_b)$, ball transfer $D(1 - \omega_b)\omega_b$ and heat $D(1 - \omega_b)^2$ gives

$$W_d = \frac{2}{5}(1 - e^{-5/2}), \quad \Delta K_b = \frac{1}{5}(1 - e^{-5/2})^2, \quad H = \frac{1}{5}(1 - e^{-5}) \quad (\text{J}). \quad (8)$$

The ball store follows independently from its final rate. With a free undriven carrier initially at rate one, the same torque laws instead give $\omega_c = 3/5 + 2e^{-25t/6}/5$, $\omega_b = 3/5 - 3e^{-25t/6}/5$ and $H = 3(1 - e^{-25/3})/25 \text{ J}$ at one second. Weighted momentum is conserved, the initial kinetic store is $3/10 \text{ J}$, and both final spin stores must be retained. A finite removal of torque preserves velocity; an impulsive release requires its own contact and transfer law.

A finite actuator model adds strain q , thermal excess H and gate voltage v_g . Declare

$$T_c = Kq + D(\omega_c - \omega_b), \quad \dot{q} = \omega_c - \omega_b, \quad (9)$$

$$I_c \dot{\omega}_c = T_d - T_c, \quad I_b \dot{\omega}_b = T_c - \omega_b/10, \quad T_d = 2(\Omega_d - \omega_c), \quad (10)$$

$$\frac{1}{100} \dot{v}_g = (V_g - v_g)/10, \quad \dot{H} = 2(\Omega_d - \omega_c)^2 + D(\omega_c - \omega_b)^2 + (V_g - v_g)^2/10 - H/2. \quad (11)$$

The independently defined store is $I_c \omega_c^2/2 + I_b \omega_b^2/2 + Kq^2/2 + H + v_g^2/200$. External powers are $T_d \Omega_d$, $\dot{K} q^2/2$, $V_g(V_g - v_g)/10$, $-\omega_b^2/10$ and $-H/2$. Multiplying the separate laws gives their sum as the store derivative. The stiffness actuator is physically distinct from the gate supply.

Use one-second stages with smooth cosine ramps $r(t) = (1 - \cos \pi t)/2$ on each local $0 \leq t \leq 1$: prepare $(\Omega_d, K, D, V_g) = (r, r/5, 1/10, 1)$; lock with $(1, 1/5, 2, 1)$; release with $(1, (1 - r)/5, 2(1 - r), 0)$; and reduce the drive with $(1 - r, 0, 1/2, 0)$. Start all states at zero. Each actual final store is evaluated from the resulting state; finite damping does not specify an exact reset. The equations define a conditional finite comparison, with no decimal trajectory asserted here. A stiffness actuator's efficiency, real release law and temperature dependence require acquisition. Both shaft channels, spring state, gate and stiffness supplies, heat export and thermal endpoints belong to that comparison.

3.4 Simple train and four observation frames

Let the sun, planet, internal ring, and carrier have indices s, p, r, c . All axes are initially parallel to $+z$, with positive right-hand rates. For common module m_0 , radii $r_j = m_0 Z_j/2$ satisfy $a = r_s + r_p = r_r - r_p$ and $Z_r = Z_s + 2Z_p$. Rolling at each mesh gives

$$\omega_p = \omega_c - \frac{Z_s}{Z_p}(\omega_s - \omega_c) = \omega_c + \frac{Z_r}{Z_p}(\omega_r - \omega_c), \quad Z_s(\omega_s - \omega_c) + Z_r(\omega_r - \omega_c) = 0. \quad (12)$$

This carrier-relative construction is consistent with [Culpepper \(2002\)](#). Three equally spaced planets additionally require $(Z_s + Z_r)/3$ integral and adequate clearance. Configuration I uses $(24, 18, 60)$ and $m_0 = 1/500 \text{ m}$, so $(r_s, r_p, r_r, a) = (3/125, 9/500, 3/50, 21/500) \text{ m}$ and the phase quotient is 28.

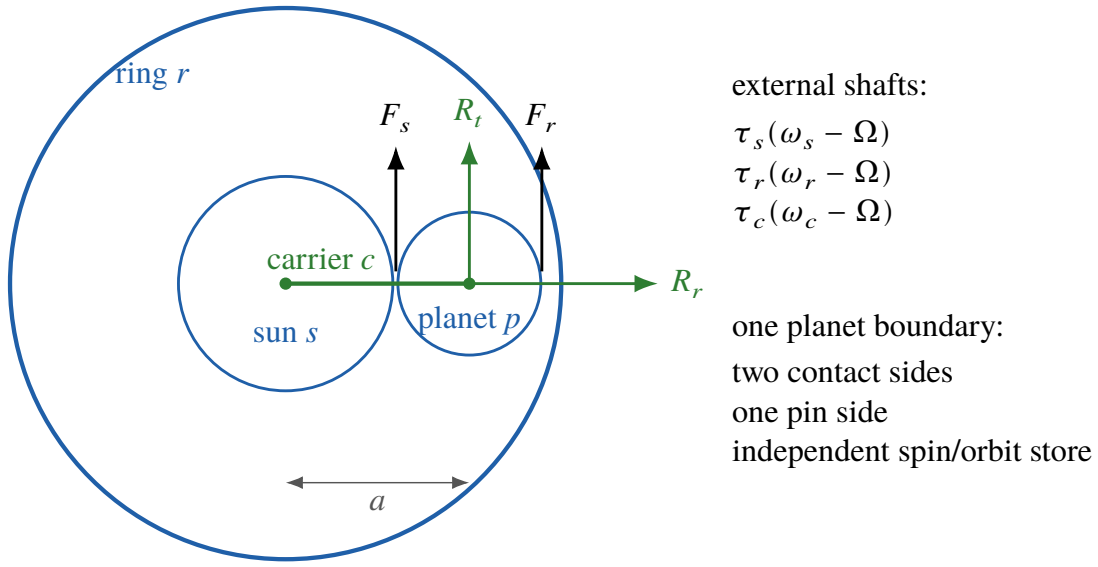


Figure 3. Schematic pitch contacts and external shaft ports. Only one planet is drawn. Force arrows indicate positive tangential and radial components; their signed values follow the free bodies.

Figure 3 locates the forces and shaft ports. For $q = 3$ equally loaded planets, tangential forces on one planet are F_s, F_r , and its pin reaction is $R_t \mathbf{e}_\theta + R_r \mathbf{e}_r$. With zero pin couple, Newton–Euler equations, independently of energy, give

$$F_s = \frac{\tau_s - I_s \dot{\omega}_s}{qr_s}, \quad F_r = F_s + \frac{I_p \dot{\omega}_p}{r_p}, \quad R_t = m_p a \dot{\omega}_c - F_s - F_r, \quad (13)$$

$$\tau_r = I_r \dot{\omega}_r + qr_r F_r, \quad \tau_c = I_c \dot{\omega}_c + qa R_t, \quad R_r = -m_p a \omega_c^2. \quad (14)$$

These radial forces belong to the tangential pitch-contact idealization. Tooth pressure angles would add radial reactions. Constant rates give $(\tau_s, \tau_r, \tau_c) = \tau_s(1, 5/2, -7/2)$ in configuration I.

The complete train contains gears, carrier, and pins. Only its three shaft ports cross the mechanical boundary in this rolling model. A planet subboundary also has the separately oriented powers

$$P_{sp}^f = F_s r_s (\omega_s - \Omega), \quad P_{rp}^f = F_r r_r (\omega_r - \Omega), \quad P_{cp}^f = R_t a (\omega_c - \Omega). \quad (15)$$

Each opposite mesh or pin side has the opposite product. Here Ω is 0, ω_c , ω_s , or ω_r for the four specified observations. Their coordinate reconstructions do not physically remount any load. The measured division among shaft, mesh and pin transfers must retain each side; a shaft work difference cannot be assigned to a pin [OP-EPI-02].

Define $J_c = I_c + qm_p a^2$, $J = I_s + I_r + J_c + qI_p$ and

$$K^0 = \frac{1}{2}(I_s \omega_s^2 + I_r \omega_r^2 + J_c \omega_c^2 + qI_p \omega_p^2), \quad H = I_s \omega_s + I_r \omega_r + J_c \omega_c + qI_p \omega_p, \quad E^f = K^0 - \Omega H. \quad (16)$$

The relative kinetic energy is $K^f = K^0 - \Omega H + J \Omega^2 / 2$; E^f includes effective centrifugal potential $-J \Omega^2 / 2$. This effective energy and the physical ground energy are distinct stores. The familiar rotating-velocity law is documented in MIT (2022).

Theorem 1 (Frame work and its path division). *For the specified rigid train, $\sum_j \tau_j = \dot{H}$ and*

$$\dot{K}^0 = \sum_j \tau_j \omega_j, \quad \dot{E}^f = \sum_j \tau_j (\omega_j - \Omega) - \dot{\Omega} H, \quad (17)$$

$$W_j^0 - W_j^f = \int \Omega \tau_j dt, \quad \int \Omega \dot{H} dt + \int H \dot{\Omega} dt = [\Omega H]_{t_0}^{t_1}. \quad (18)$$

Proof. Sum (14) after multiplying the spin and translation equations by their own rates. The opposite contact products cancel using (12); summing moments uses both equalities defining a . Differentiating (16), subtracting each shaft product, and integrating the product derivative of ΩH gives the remaining identities. $\square$

The effective-field term separates as Euler power $-\dot{\Omega}(H - J\Omega)$ and explicit potential-time term $-J\Omega\dot{\Omega}$. Each is integrated before combining them. They are coordinate terms, not physical motor or heat supplies. Reconstructing these transient frame balances within independent uncertainty is the physical comparison [OP-EPI-01]. For a genuine rotating support its physical torque–rate work is additional.

For exact rational examples choose $I_s = I_r = qI_p = J_c = 1 \text{ kg m}^2$, $m_p = 1 \text{ kg}$, and $I_c = 248677/250000 \text{ kg m}^2 > 0$. Flywheels may supply these inertias; no uniform-disc mass law is assumed. Let $(\omega_s, \omega_r, \omega_c, \omega_p) = (7/2, 0, 1, -7/3) \text{ rad/s}$ and $\tau_s = 2 \text{ N m}$ on $[0, 1] \text{ s}$. Direct state evaluation gives $H = 13/6$, $K^0 = 673/72$, $E^c = 517/72$ in SI units at **both** endpoints.

Table 3. Configuration I in steady operation. Only the first three entries belong to the complete external shaft sum. Each complete residual is zero.

Separately integrated work, J	Ground	Carrier
Sun shaft	7	5
Ring shaft	0	−5
Carrier shaft	−7	0
One sun contact into planet	7/3	5/3
One ring contact into planet	0	−5/3
One pin into planet	−7/3	0

Preparation instead prescribes $\omega_j(t) = \omega_{j*}t$ and $\tau_s(t) = 2t$ on $[0, 1] \text{ s}$ from rest. Equations (14) give $\tau_r = 5t - 595/36$ and $\tau_c = 673/36 - 7t$. For any frame $\Omega = \Omega_*t$, each shaft integral is

$$W_j^f = (\omega_{j*} - \Omega_*) \left(\frac{a_j}{3} + \frac{b_j}{2} \right), \quad (a_s, a_r, a_c) = (2, 5, -7), \quad (b_s, b_r, b_c) = (0, -595/36, 673/36). \quad (19)$$

The units of a_j, b_j here incorporate the one-second ramp. Every internal integral follows (15) with the affine forces in (14); explicitly $\int_0^1 (At + B)t dt = A/3 + B/2$.

Table 4. Preparation works and independent endpoints. Every complete sum equals its store change. The ground and ring frames coincide only for this held-ring path.

Observation	$(W_s, W_r, W_c), \text{ J}$	Effective-field work,	
		J	$E(0), E(1), \text{ J}$
Ground and held ring	$(7/3, 0, 505/72)$	0	$0, 673/72$
Carrier	$(5/3, 475/72, 0)$	$-13/12$	$0, 517/72$
Sun	$(0, 3325/144, -2525/144)$	$-91/24$	$0, 127/72$

For a distinct history set $H = (13/6)t$ and $\Omega = t + bt(1-t)$. With $d = \omega_s - \omega_c$ one has $H = 4\omega_c - (11/15)d$, so choose $\omega_c = \Omega$ and $d = (15/11)(4\Omega - H)$. Equation (12) determines all other rates; the ring moves during preparation when $b \neq 0$ and reaches the same held endpoint. The shaft and field differences are $(13/6)(1/2 + b/6)$ and $(13/6)(1/2 - b/6)$ J. Thus $b = 0, 1$ give respectively $(13/12, 13/12)$ and $(13/9, 13/18)$ J. The same endpoint total $13/6$ J follows by integration by parts, not by presuming an equal division or a held ring throughout a nonproportional preparation. Measuring this history-dependent shaft/field division is a separate comparison [OP-EPI-04].

For a locked approach retain $\omega_c = 1$, put $\omega_s = 1 + \epsilon$, and obtain $\omega_r = 1 - 2\epsilon/5$, $\omega_p = 1 - 4\epsilon/3$. At fixed $\tau_s = 2$ the one-second ground shaft works are $(2 + 2\epsilon, 5 - 2\epsilon, -7)$ J. At $\tau_s = 2\epsilon$ they are ϵ times that triple. Their exact limits are $(2, 5, -7)$ and $(0, 0, 0)$ J. Carrier-frame works vanish at the locked endpoint in both controls. The independent kinetic endpoints are evaluated by (16); a steady interval has identical states at its ends, not necessarily zero energy. Fixed-drive and scaled-drive approaches require their own finite uncertainty bounds [OP-EPI-12].

3.5 Rotor, stator, phase, and load

Let $x = \theta_p - \phi$, where $\dot{\phi} = \omega_c$, and specify $\theta_o = \phi + F(x)$. The output rate is $\omega_o = \omega_c + \chi(\omega_p - \omega_c)$, $\chi = F'(x)$. For the same rotor path two stator counts obey $N_o^0 - N_o^c = \Delta\phi/(2\pi)$. Physical stator remounting under load requires a fresh trajectory, not substitution into the unloaded path.

For a Hooke joint define the continuous angle lift $H_\beta(x) = \text{atan2}(\sin x, \cos \beta \cos x)$. The two-joint Z path is $F = H_{\beta_2}(H_{\beta_1}(x) + \gamma + \pi/2) + \pi/2$, up to a constant. Equal angles and carrier phasing $\gamma = 0$ give $F = x$; quarter-turn phasing gives

$$\tan F = \frac{\tan x}{\cos^2 \beta}, \quad \chi = \frac{\cos^2 \beta}{\cos^4 \beta \cos^2 x + \sin^2 x}, \quad F(x + \pi) = F(x) + \pi. \quad (20)$$

An arbitrary finite readout retains $[F(x) - x]_{t_0}^{t_1}/(2\pi)$. Ground phasing prescribes $\gamma = -\phi$ in the full map and introduces its own phase actuator; it supplies a mean carrier-relative count with endpoint ripple. The paired-stator identity does not prohibit differential instruments that report other relative observables. These maps cover the eleven specified ideal mountings: equal-angle correct and quarter-turn Hooke paths against ground and carrier stators; the kinematic ground-phased Hooke path with ground stator; constant-ratio offset paths against both stators; and bevel ratios one and two against both stators. Oldham's orthogonal sliding coordinates are $a \cos x, a \sin x$. A Schmidt path can share the angle map without sharing its force system.

For configuration II the linkage contains input inertia I_R , orbiting mass m_R , output inertia I_o , and

massless joint members. An imposed output torque τ_L and drag $d_0 \operatorname{sgn}(\omega_o - \omega_\sigma)$ give

$$\mathcal{T} = I_o \dot{\omega}_o - \tau_L + d_0 \operatorname{sgn}(\omega_o - \omega_\sigma), \quad \tau_d = I_R \dot{\omega}_p + \chi \mathcal{T}, \quad \tau_u = (1 - \chi) \mathcal{T}. \quad (21)$$

The five physical work channels are the separate integrals of $\tau_d \omega_p$, $\tau_u \omega_c$, $\tau_L \omega_o$, $-d_0 \operatorname{sgn}(\omega_o - \omega_\sigma) \omega_o$, and $m_R a^2 \dot{\omega}_c \omega_c$. Their sum differentiates $K_L^0 = (I_R \omega_p^2 + m_R a^2 \omega_c^2 + I_o \omega_o^2)/2$. The stator reaction has power $+d_0 \operatorname{sgn}(\omega_o - \omega_\sigma) \omega_\sigma$; including it gives mechanical conversion $D = d_0 |\omega_o - \omega_\sigma|$. Thermal storage changes the larger boundary as derived below.

For constant input rates, constant load, and unchanged drag sign, the Hooke support-cycle integral over $T_b = \pi/|\omega_p - \omega_c|$ is zero: $\int (1 - \chi) dt = 0$ and its inertial part is proportional to $[\chi - \chi^2/2]$ at equal phases. In a massless control set $(\omega_p, \omega_c) = (3, 1)$, $\tau_L = -1$, $d_0 = 0$. On $[0, \pi/2]$ s the Hooke drive, support, and receiver works are $(3\pi/2, 0, -3\pi/2)$ J. The bevel map $F = 2x$ instead gives $(3\pi, -\pi/2, -5\pi/2)$ J. Both ideal endpoint stores are zero. In the carrier frame the bevel works are $(2\pi, 0, -2\pi)$ J. A whole-cycle support null cannot be inferred for a partial cycle. Actual Hooke support work is the first measurement [OP-EPI-06]; the different bevel support path needs its own work account [OP-EPI-07].

For unequal angles with correct phasing, $\tan F = (\cos \beta_2 / \cos \beta_1) \tan x$. Choose these cosines as $3/5, 4/5$ respectively, in the order $\cos \beta_2 = 3/5$, $\cos \beta_1 = 4/5$. With $x = t$, $\omega_c = 1$, $\mathcal{T} = 1$, and $[0, \pi/4]$ s, the support work is $\pi/4 - \arctan(3/4) > 0$ J, against zero for equal angles. These deliberately large articulations are mathematical controls; buildability and clearance require separate geometry. Under carrier acceleration, the rotor-pin integral for $m_R a^2 = 1$ and $\omega_c = t$ is $1/2$ J. For an independently moving phase coordinate γ , virtual displacement adds torque $\tau_\gamma = \mathcal{T} \partial \theta_o / \partial \gamma$; its work is $\int \tau_\gamma \dot{\gamma} dt$. A unit torque through $\gamma = t$ supplies 1 J. None is supplied by a count alone. Their combined physical lead-out, carrier acceleration and phase-reference works remain to be determined [OP-EPI-31].

The actual Cardan force and bending system requires its bearing constraints, member stiffnesses, and application-point motions. A finite control with $F_x = 1$ N, $\dot{x} = 1$ m/s over one second transfers 1 J; a fixed support transfers zero at the same force. The same rule applies to a bending couple. It specifies a force/displacement discrimination, not the unknown force in an unspecified double Cardan device. Full transverse forces and bending couples, including their application-point motion, are the acquisition target [OP-EPI-30].

For a stator comparison reproducing the finite scale of Nilre and Herlin (2026), take $\omega_p = -14\pi/3$, $\omega_c = 2\pi$, $d_0 = 1/50$ and duration $7/2$ s. The separate relative drag integrals are $Q_0 = 49\pi/150$ and $Q_c = 7\pi/15$ J, giving $Q_c - Q_0 = 7\pi/50$ J. This is a change of physical mounting, not a coordinate dependence of the heat produced by one mounting. The actual loaded readout and its finite endpoint corrections require identification [OP-EPI-05].

3.6 Attaching a receiver changes the free bodies

Attach three identical unity-ratio, ground-stator lead-outs to configuration I, with $\tau_L = +1$ N m on the negative-speed outputs and zero drag. The steady linkage requires $\tau_d = -1$, $\tau_u = 0$. The planet equation becomes $F_r = F_s + (I_p \dot{\omega}_p + \tau_d)/r_p$; the carrier equation also includes $q(\tau_u + m_R a^2 \dot{\omega}_c)$. The resulting steady torques are $(2, -5, 0)$ N m. Separate one-second external works are

$$(W_s, W_r, W_c) = (7, 0, 0) \text{ J}, \quad W_{L,j} = -7/3 \text{ J} \quad (j = 1, 2, 3). \quad (22)$$

The shared axle transfers $+7/3$ J to each linkage and $-7/3$ J to its planet. Both endpoint stores are $673/72 + 3[(I_R + I_o)(7/3)^2 + m_R a^2]/2$ J for the stated added inertias. They agree because the actual

endpoint rates agree. The carrier work changes from -7 to zero, a $+7$ J comparison. Adding a receiver to an unchanged unloaded force table would miss that change. The full joint, bearing, phase-driver, and thermal balance still requires those independent physical ports and states. The corresponding complete loaded-train work remains a physical measurement [OP-EPI-33].

4 Additional mechanical boundaries

4.1 Compound circulation and a distinct two-sun geometry

In configuration III two coaxial members A, B engage rigidly connected planet sections P_A, P_B carried by C . Let $s_X = -1$ for an external mesh, $+1$ for an internal mesh, and

$$a = \frac{m_0}{2}(Z_A - s_A Z_{pA}) = \frac{m_0}{2}(Z_B - s_B Z_{pB}), \quad h_X = s_X \frac{Z_{pX}}{Z_X}, \quad R = \frac{h_A}{h_B}. \quad (23)$$

Rolling gives $\omega_X - \omega_C = h_X(\omega_P - \omega_C)$. The planet spin equation $s_A r_{pA} F_A + s_B r_{pB} F_B = 0$, together with shaft and carrier free bodies, gives $(\tau_A, \tau_B, \tau_C) = \tau_*(1, -R, R - 1)$. This is a massless quasistatic model with empty endpoint stores.

Three shaft powers and opposite sides of the two contacts and pin give nine component powers, namely signed copies of $p_X^f = \tau_X(\omega_X - \Omega)$. Define $D_0 = \max_X |\tau_X \omega_X| > 0$. Then

$$\mathcal{E}_f = \frac{\max_X |p_X^f|}{D_0}, \quad \mathcal{E}_0 = 1, \quad \mathcal{E}_C = \begin{cases} 1/|R - 1|, & A \text{ held,} \\ |R|/|R - 1|, & B \text{ held.} \end{cases} \quad (24)$$

Each work is separately $p_X^f T$, with its opposite side integrated as $-p_X^f T$ before combining. Choose $\omega_C = 1$, $\tau_* = 1$, $T = 1$ in SI units. Five pitch-compatible tooth sets give the following exact values.

Table 5. Compound controls. Ground ratios are one in all cases. The other frames follow by putting $\Omega = \omega_A$ or ω_B in equation (24). Multiple-planet phasing is a separate assembly condition; a single stepped planet suffices for these kinematic controls.

$(Z_A, Z_{pA}, Z_{pB}, Z_B)$	(s_A, s_B)	R	$\mathcal{E}_C, A \text{ held}$	$\mathcal{E}_C, B \text{ held}$
(24, 18, 18, 60)	$(-1, +1)$	$-5/2$	$2/7$	$5/7$
(20, 22, 20, 62)	$(-1, +1)$	$-341/100$	$100/441$	$341/441$
(30, 20, 19, 31)	$(-1, -1)$	$62/57$	$57/5$	$62/5$
(62, 20, 21, 63)	$(+1, +1)$	$30/31$	31	30
(100, 58, 59, 101)	$(+1, +1)$	$2929/2950$	$2950/21$	$2929/21$

At $R = 1$ with either coaxial member held, $D_0 = 0$; the ratio is undefined. No finite resolution on the denominator establishes an infinite physical ratio. Approaching that geometry at finite tooth numbers is a sequence of specified comparisons. Their frame-dependent power ratios require synchronized torque and speed measurements with a resolved denominator [OP-EPI-08].

A Ravigneaux arrangement has independent long and short planet spins, not the single P of (23). To exhibit the distinction, use two sun planes and one long/short pair with teeth $(Z_A, Z_B, Z_R, Z_S, Z_L) = (30, 60, 90, 15, 15)$. With $u_j = \omega_j - \omega_C$, its four independent rolling equations are

$$30u_A + 15u_S = 0, \quad 15u_S + 15u_L = 0, \quad 60u_B + 15u_L = 0, \quad 90u_R - 15u_L = 0. \quad (25)$$

At module $1/30$ m the short and long centres can lie on the same radial line at $3/4$ and $5/4$ m; their separation is $1/2$ m, the sum of their pitch radii. Sun planes are axially separated. This is pitch geometry, not a tooth-strength or packaging design. With $\omega_C = 1$, $\omega_R = 0$, rates are $(\omega_A, \omega_B, \omega_S, \omega_L) = (-2, 5/2, 7, -5)$. The independent reduced constraints are $u_A + 2u_B = 0$ and $u_A - 3u_R = 0$. Their massless contact reactions have multiplier torque vectors $(1, 2, 0, -3)$ and $(1, 0, -3, 2)$ on (A, B, R, C) , as follows by resolving each contact's equal and opposite force. Choosing $\tau_A = \tau_B = 1$ gives $\tau_R = -3/2$, $\tau_C = -1/2$. The separate one-second ground shaft works are $(-2, 5/2, 0, -1/2)$ J; the carrier works are $(-3, 3/2, 3/2, 0)$ J. Both sum to zero, and the shaft ratio is $6/5$ against its ground denominator $5/2$ W. Internal contact powers require these four meshes, not the earlier nine-side set. Empty massless stores close this bounded model only.

Helical thrust or a nonparallel joint adds vector contact and support ports. For instance an imposed axial force 1 N through 1 m over one second adds 1 J, whereas holding that coordinate gives zero. Its existence and sign in a given gear require that gear's contact geometry, compliance, and measured trajectory. The two-sun example supplies no such missing law. Other geometry and contact directions retain their own energy partition [OP-EPI-21].

4.2 Contact changes, reversals, and unequal loading

A compliant gap has overlap δ , stored energy $U = k[\max(\delta, 0)]^2/2$, and, during compression contact, $F = k\delta + c\dot{\delta}$. Keep the damper active only on its declared contact branch; separating and tensile branches need their own force law. For the spring/damper subboundary, input is $F\dot{\delta}$ and heat export is $-c\dot{\delta}^2$. At $\delta = 0$ the spring store is continuous even when damping makes power jump. The control $\delta = t$, $k = 2$, $c = 1$ on $[0, 1]$ s has

$$W_{\text{contact}} = \int_0^1 (2t + 1)dt = 2 \text{ J}, \quad W_h = -1 \text{ J}, \quad U(0) = 0, \quad U(1) = 1 \text{ J}. \quad (26)$$

At onset input power changes from 0 to 1 W, while ΔU across the instant itself and impulse work are zero. Ramps approaching contact are separate intervals; the actuator imposing δ has the opposite work. Actual re-engagement power jumps require the finite contact law and event-side measurements [OP-EPI-10].

For an impulsive alternative take two freely translating effective masses m_1, m_2 and relative incoming speed $v_- = v_1^- - v_2^- > 0$. A declared restitution coefficient $e \in [0, 1]$ gives $J = (1 + e)\mu v_-$, $\mu = m_1 m_2 / (m_1 + m_2)$, from impulse momentum and $v_+ = -e v_-$. The separate contact-side works under a resolved regularization with no other impulse are

$$W_1 = \frac{m_1}{2}[(v_1^+)^2 - (v_1^-)^2], \quad W_2 = \frac{m_2}{2}[(v_2^+)^2 - (v_2^-)^2], \quad W_1 + W_2 = -\frac{\mu}{2}(1 - e^2)v_-^2. \quad (27)$$

These follow directly by integrating $m\dot{v}$, not by multiplying a force impulse by one arbitrarily chosen event velocity. For unit masses, $(v_1^-, v_2^-) = (1, 0)$ and $e = 1/2$, final velocities are $(1/4, 3/4)$, $J = 3/4$ N s, works $(-15/32, 9/32)$ J, and kinetic stores $1/2, 5/16$ J. The contact receives $3/16$ J. If contact and its thermal store are included, that conversion is internal; if its heat is exported, the external heat work is $-3/16$ J. Do not also add the two internal contact works to that larger boundary. A finite pulse with the specified restitution can measure these transfers; it cannot realize a zero-time impulse. Physical contact regularization and its energy destination remain independent questions [OP-EPI-22].

For a full compliant train write actual contact coordinates $\delta_l(q)$, $U = \sum_l k_l \max(\delta_l, 0)^2/2$, and apply each force through its geometric gradient. The rigid no-slip constraints must be released where that compliance acts. Multiplying the full Newton equations by all body velocities gives $\dot{K} + \dot{U} = \sum P_{\text{ext}} - \sum_l c_l \dot{\delta}_l^2$ on each

active contact interval. In a coaxial observation add $-\dot{\Omega}H$ to the effective energy $K + U - \Omega H$. An explicit full-train control includes all three planets and both contacts on each. Keep configuration I's inertias, hold the sun, ring, and planet spin angles at zero, and prescribe $\theta_c = t^2/2$ on $[0, 1]$ s. Define the two linear pitch-compliance coordinates on each planet by $\delta_s = a\theta_c - r_s\theta_s - r_p\theta_p - g$ and $\delta_r = a\theta_c - r_r\theta_r + r_p\theta_p - g$. Let $g = a/8$, $k = 2/a^2$, and $c = 1/a^2$ in SI units. All six contacts engage at $t = 1/2$ s; during contact they have $\delta = a(t^2 - 1/4)/2$, $\dot{\delta} = at$, and $F = (t^2 + t - 1/4)/a$. Equal opposite planet spin torques leave $\omega_p = 0$ without a planet drive. Resolving each contact gradient gives $\tau_s = -3r_sF$, $\tau_r = -3r_rF$, and $\tau_c = 1 + 6aF$ after engagement; before it, $(\tau_s, \tau_r, \tau_c) = (0, 0, 1)$. This is an exact lumped linear pitch-contact model, not a claim about finite-deflection tooth geometry.

Separate integration on the gap and contact intervals gives

$$W_c^0 = 99/32, \quad W_s^0 = W_r^0 = 0, \quad \sum_{l=1}^6 W_{h,l} = -7/4, \quad (K+U)(0) = 0, \quad (K+U)(1) = 43/32 \quad \text{J.} \quad (28)$$

Each heat work is $-7/24$ J; each spring endpoint is $9/64$ J and final carrier kinetic energy is $1/2$ J. In carrier observation, $(W_s^c, W_r^c, W_c^c) = (83/112, 415/224, 0)$ J, field work is $-1/2$ J, and effective endpoints are $0, 11/32$ J. At engagement damping makes finite contact powers jump; coordinates, velocities, and stores remain continuous and impulse work is zero. These independently evaluated stores close both balances. Clearance in a lead-out, changing active tooth contacts, unequal loading, and actual force laws remain separate physical questions. Their full accelerating-train remainder is not fixed by this contact control [OP-EPI-23].

With unequal planet fractions $(1/2, 1/3, 1/6)$, a steady total 7 J sun-contact transfer divides into $(7/2, 7/3, 7/6)$ J rather than three $7/3$ J values. The fractions specify a load-sharing law; they are not inferred from an aggregate balance. A pin couple $M = 1/10$ N m on a planet with $\omega_p - \omega_c = -10/3$ rad/s gives pair work $-1/3$ J in one second. Its two separately integrated sides are $M\omega_p$ and $-M\omega_c$. Bearing heat or elastic torsion then needs an independently chosen law. Each planet's force and couple channels must therefore be observed. The load-sharing and pin-couple partition is a local measurement [OP-EPI-24].

For a reversal, $\omega = 1 - 2t$, $\tau = 1$ on $[0, 1]$ s gives $W_{[0,1/2]} = 1/4$ J and $W_{[1/2,1]} = -1/4$ J. Net work is zero while $\int |P|dt = 1/2$ J. A separate Coulomb drag of unit magnitude converts $\int |\omega|dt = 1/2$ J; treating its sign as constant would give the wrong zero. A unit inertial rotor has equal endpoint kinetic energies $1/2$ J on this prescribed reversal, and its inertial actuator torque $\dot{\omega} = -2$ supplies works $-1/2, +1/2$ J on the two halves. Any concurrent applied constant torque and brake require their own compensating drive. The examples are separate effort laws, not a claim that torque one alone produces that trajectory. Signed and absolute works through an actual reversal need independently calibrated channels [OP-EPI-28].

4.3 Radial motion and vector observation

For a centre at radius r , planar Newton equations give $F_r = m(\ddot{r} - r\omega_c^2)$ and $F_t = m(r\dot{\omega}_c + 2\dot{r}\omega_c)$. Both $F_r\dot{r}$ and $F_t r\omega_c$ are physical work channels. With $m = 1$, $r = 1 + t$, $\omega_c = 1$ over one second, their works are $-3/2$ and $+3$ J. Ground kinetic energy changes from 1 to $5/2$ J; carrier effective energy changes from 0 to $-3/2$ J. This radial-guide control requires compliant or variable contact geometry before it can represent an actual radially moving planet.

In any rotating observation, $\mathbf{F}_{\text{Cor}} = -2m\boldsymbol{\Omega} \times \mathbf{v}^f$ has $\mathbf{F}_{\text{Cor}} \cdot \mathbf{v}^f = 0$. At fixed radius, $\mathbf{v}^f = r(\omega_c - \Omega)\mathbf{e}_\theta$, so the Coriolis force is generally nonzero outside ground and carrier observations. Radial motion adds a tangential component; it does not add net Coriolis power. This distinction follows from the vector product

itself. Radial support work and changed contact geometry still need identification [OP-EPI-26].

For observations rotating about a fixed common origin in three dimensions,

$$E^f = K^0 + U - \mathbf{\Omega} \cdot \mathbf{H}, \quad P_j^f = \mathbf{F}_j \cdot (\mathbf{v}_j - \mathbf{\Omega} \times \mathbf{r}_j) + \mathbf{M}_j \cdot (\boldsymbol{\omega}_j - \mathbf{\Omega}), \quad P_{\text{field}} = -\dot{\mathbf{\Omega}} \cdot \mathbf{H}. \quad (29)$$

Here U is rotationally invariant internal potential, and all vectors and time derivatives in the equation are resolved in an inertial basis. The result follows from $\sum (\mathbf{r} \times \mathbf{F} + \mathbf{M}) = \dot{\mathbf{H}}$ and the kinetic-energy identity, including the full inertia tensor in the relative kinetic and centrifugal terms. A translating origin would add linear-momentum terms and is outside this formula. For a free spherical rotor of unit inertia with $\boldsymbol{\omega} = (1, 0, 2)$ and observation $\mathbf{\Omega} = (t, 0, 0)$, $K^0 = 5/2$ J at both endpoints, physical work is zero, field work is -1 J, and effective energy changes from $5/2$ to $3/2$ J. Dropping the transverse momentum predicts zero field work. A symmetric axial train can have zero transverse momentum; the rotor example distinguishes the vector laws without asserting that every train has this nonzero component. The nonparallel-axis energy comparison requires the full observed vector momentum [OP-EPI-27].

5 Windings, returns, and finite dynamic correspondence

5.1 Coordinate zero versus an attached branch

In a tapped pair, currents enter winding 1 at a and winding 2 at b , with returns c and a . Thus

$$(v_1, v_2) = (V_a - V_c, V_b - V_a), \quad (I_a, I_b, I_c) = (i_1 - i_2, i_2, -i_1). \quad (30)$$

A common $V_j \mapsto V_j + g(t)$ leaves the complete winding voltages unchanged. Its terminal contributions change by gI_j , whose sum is zero by current conservation. Each changed work is separately $\int gI_j dt$; deleting a return changes that sum.

A physical attachment instead has voltage v_b across its own resistor R_b and capacitor C_b . On a prescribed ramp $v_b = t$ with $R_b = C_b = 1$ over one second, its source current is $i_b = t + 1$. For the attachment boundary, direct integration gives

$$W_b = \int_0^1 t(t+1)dt = 5/6 \text{ J}, \quad W_h = -\int_0^1 t^2 dt = -1/3 \text{ J}, \quad E_b(0) = 0, \quad E_b(1) = 1/2 \text{ J}. \quad (31)$$

Adding a parallel constant-current receiver of 1 A over the same ramp gives another $1/2$ J of receiver delivery and raises source work to $4/3$ J. More generally $W_b = C_b/2 + 1/(3R_b)$ J for the unit ramp; a pure voltage coordinate change gives no added branch work or capacitor store. The driver holding that voltage must be included in the larger circuit; without it the attached branch changes the voltage trajectory. Ground-bond impedance, probe common-mode paths, interwinding capacitance, and chassis capacitance are distinct branches with their own endpoints. Their physical attachment work is the return question [OP-TRF-01].

5.2 A regular winding model and its seven ports

Configuration IV has reciprocal inductance

$$\mathbf{L} = \begin{pmatrix} L_1 & M \\ M & \rho^2 L_1 \end{pmatrix}, \quad M = sk\rho L_1, \quad s = \pm 1, \quad 0 \leq k < 1, \quad E_m = \frac{1}{2}i^T \mathbf{L}i, \quad E_C = \frac{1}{2}Cv_o^2. \quad (32)$$

Here L_1, C, ρ are positive. The determinant is $\rho^2 L_1^2 (1 - k^2) > 0$. An exact decomposition is $E_m = k L_1 (i_1 + s \rho i_2)^2 / 2 + (1 - k) L_1 i_1^2 / 2 + (1 - k) \rho^2 L_1 i_2^2 / 2$. The distinction between finite magnetic storage and the ideal transformer is standard in [Haus and Melcher \(1989\), Section 9.7](#).

Let $\eta = 1$ for tapped and 0 for isolated connections, whose returns remain physically distinct. A source u behind $R_s > 0$, a core shunt G_c , and output load/probe conductances G_L, G_P give, by Kirchhoff laws,

$$v_a = \frac{u - R_s(i_1 - \eta i_2)}{1 + R_s G_c}, \quad I_s = i_1 - \eta i_2 + G_c v_a, \quad (33)$$

$$L\dot{i} = (v_a, v_o - \eta v_a)^T - \text{diag}(R_1, R_2)i, \quad C\dot{v}_o = -i_2 - (G_L + G_P)v_o. \quad (34)$$

The boundary includes windings, output capacitor, and the source, copper, and core resistors. The source, receiver, probe, and immediate heat reservoirs are external. Its seven separate powers are

$$(uI_s, -R_s I_s^2, -R_1 i_1^2, -R_2 i_2^2, -G_c v_a^2, -G_L v_o^2, -G_P v_o^2). \quad (35)$$

Each negative resistor product is its voltage times outward current or, under immediate export, $-T_b \dot{S}_{\text{out}}$. Multiplication of (34) by i^T and v_o derives $\dot{E}_m + \dot{E}_c = \sum P_j$. This does not assume that either store is constant. Equation (2) gives each work from its own quadratic form, and endpoints use (32) on the propagated state.

Use $L_1 = 1/25$ H, $\rho = 1$, $k = 19/20$, $s = +1$, $C = 1/500$ F, $R_s = 1/5$ Ω , $R_1 = R_2 = 1/10$ Ω , $G_c = 1/100$ S. A finite crossed comparison retains all combinations

$$G_L \in \{1/10, 2/5\}, \quad G_P \in \{0, 1/10\}, \quad \omega \in \{0, 2\pi\}, \quad (36)$$

$$x(0) \in \{(0, 0, 0), (0, 0, 1)\}, \quad u(t) = \cos(\omega t) \text{ V}.$$

Conductances are in siemens and states in amperes, amperes, volts. On $[0, 1]$ s this specifies 16 regular initial-value problems for each chosen connection. Preparation of the charged state is its own interval. For every choice, set $x = (i_1, i_2, v_o)$ and construct A directly from (34), adding $\dot{c} = -\omega s$, $\dot{s} = \omega c$ with $(c, s) = (1, 0)$. All seven predictions are the closed forms $W_j = \mathcal{T}(Q_j, A, y_0, 1)$, and both endpoint stores are specified by (32); this is an exact finite family, not a numerical trajectory. Changing the load or source at a finite event keeps i, v_o continuous when a current path remains. Reinitialize the source coordinates on the correct event side and integrate the next interval separately.

A simpler probe branch displays an explicit nonzero contrast. A capacitor $C = 1$ F initially at 1 V discharges with source disconnected through $G_L = 1$ S. With no probe its receiver work at $T = 1$ is $-(1 - e^{-2})/2$ J and endpoint store $e^{-2}/2$ J. With $G_P = 1$ S they become $-(1 - e^{-4})/4$ J and $e^{-4}/2$ J; the probe has its own work $-(1 - e^{-4})/4$ J. These follow by separately integrating $-G_j v^2$, $v = e^{-(G_L + G_P)t}$. The winding comparison in (36) still requires its magnetic states and actual source.

Moving the tapped receiver return requires another current law. Keep the capacitor across b – c and let $h = 1$ return the receiver to a , $h = 0$ to c . With $G = G_L + G_P$, $v_L = v_b - h v_a$,

$$v_a = \frac{u - R_s(i_1 - i_2) + R_s h G v_b}{1 + R_s(G_c + hG)}, \quad I_s = i_1 - i_2 + G_c v_a - h G v_L, \quad C\dot{v}_b = -i_2 - G v_L. \quad (37)$$

The two winding equations use voltages $(v_a, v_b - v_a)$. Both receivers have their own separately prepared trajectories and $-G_L v_L^2$ work. At $u \neq 0, x = 0$, return a has nonzero initial receiver power and return c zero. At $u = 0, x = (0, 0, V)$ the return- a voltage is $V(1 + R_s G_c)/(1 + R_s(G_c + G))$, smaller in magnitude

than V . Continuity preserves each strict comparison on a finite following interval. There is therefore no preparation-independent ordering of the two works. Their numerical values for a specified duration remain the separate matrix primitives, not a ratio of voltage integrals. The crossed physical load, probe, drive and preparation comparison remains open [OP-TRF-06].

5.3 Opening, commutation, and separate constrained limits

Physical commutation can connect two receiver branches, G_a across $b-a$ and G_o across $b-c$, during overlap. Replace the last equation of (37) by $C\dot{v}_b = -i_2 - G_a(v_b - v_a) - G_o v_b$ and use $v_a = [u - R_s(i_1 - i_2) + R_s G_a v_b] / [1 + R_s(G_c + G_a)]$. The two receiver powers are separately $-G_a(v_b - v_a)^2$ and $-G_o v_b^2$. A finite piecewise-constant selector, with duration 1/10 s in each of the states $(G_a, G_o) = (1, 0), (1, 1), (0, 1)$ S, has exact matrix works on all three intervals. An open-gap alternative uses $(0, 0)$ in its middle interval and needs its own propagated states. Binary h is not continuously interpolated to represent these connections.

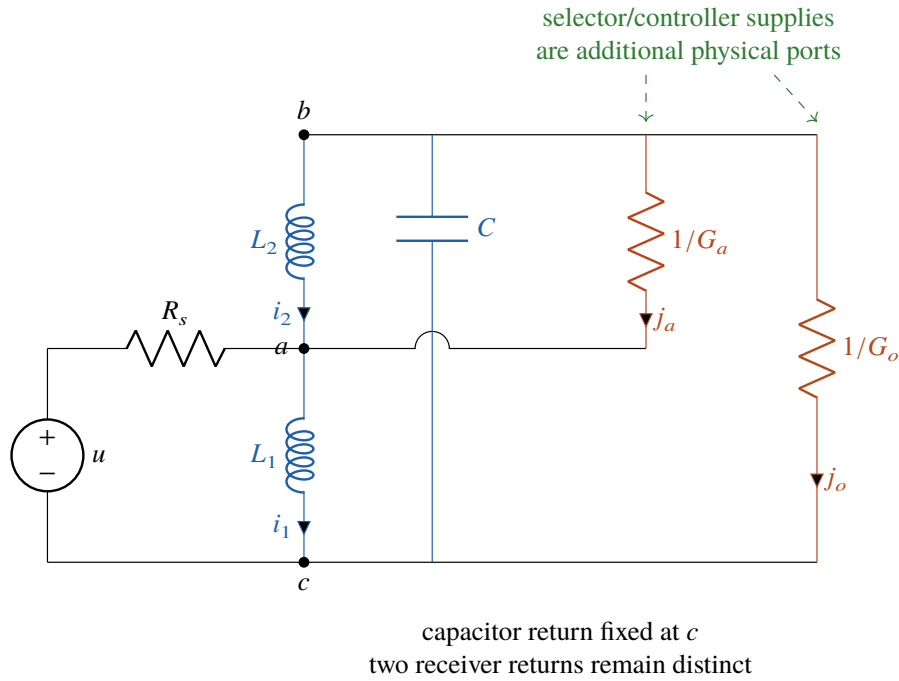


Figure 4. Schematic tapped receiver branches during commutation. Winding copper and core-loss elements are suppressed in this connection drawing and retained in the equations. The bridge at the capacitor crossing is not a junction. The two inductors are magnetically coupled.

Figure 4 keeps the two receiver branches distinct. The switch/controller boundary adds $v_{\text{ctrl}} i_{\text{ctrl}}$, its own electric and thermal states, and any mechanical actuation. Actual winding opening also adds clamp, arc, and displacement-current paths; a source command of zero with R_s still connected is a different event. As a finite clamp control, an isolated $L = 1$ H inductor prepared at 2 A drives a constant opposing clamp voltage 1 V over $[0, 1]$ s. Its law $\dot{i} = -1$ gives

$$W_{\text{clamp}} = -\int_0^1 (2 - t) dt = -3/2 \text{ J}, \quad E_L(0) = 2 \text{ J}, \quad E_L(1) = 1/2 \text{ J}. \quad (38)$$

The clamp receiver gets 3/2 J. Releasing at current 1 A into a declared parallel capacitor with zero initial

voltage permits a subsequent LC interval; simply deleting that current is not a specified transfer. A switch arc with constant voltage 2 V on the same initial inductor instead reaches zero current in one second and receives 2 J. These are two different finite laws, each with its own vi integral and endpoint.

Singular models must be written before their finite approaches:

Table 6. Configuration IV boundary models. The units of the explicit unit controls are SI. An algebraic constraint with incompatible preparation needs a named parasitic or supply path; it cannot receive an assigned impulse from this table.

Exact condition	Constrained law and prepared-state issue	Exact finite contrast
$k = 1$	$q = i_1 + s\rho i_2, L_1 \dot{q} = v_1 - R_1 i_1;$ $v_2 - R_2 i_2 = s\rho(v_1 - R_1 i_1)$	$L_1 = \rho = 1$, zero copper and $v_1 = v_2 = 1$ give $q = t$, $E_m(1) = 1/2$ J; a zero-store ideal transformer gives 0
$C = 0$	$i_2 + Gv_o = 0$; no independent capacitor state	$C = \epsilon, v_o = 1$ gives $E_C = \epsilon/2$; $v_o = \epsilon^{-1/2}$ gives $1/2$ J
$R_s = 0$	$v_a = u$; source current follows KCL	A capacitor ramp $v = t$ has $W_s = C/2 + R_s C^2$ and heat $-R_s C^2$ J on one second
Exact output short	$v_o = 0$; nonzero prepared voltage is incompatible without an edge law	A resistor across $C = 1, V = 1$ for $R \ln 2$ seconds exports $3/8$ J and leaves $1/8$ J
$\rho = 0$	$L_2 = M = 0$; winding 2 becomes $v_2 = R_2 i_2$, or $v_2 = 0$ if also $R_2 = 0$	With $L_1 = 1, i_1 = 0, i_2 = 1/\rho$ the finite pair retains $E_m = 1/2$ J

For the perfect-coupling ramp, imposing $i_2 = 0$ gives $i_1 = t$; the separately integrated primary input is $1/2$ J and secondary input zero. Thus the example includes its effort allocation as well as q . For $C = 0$ or zero turns ratio, bounded-state approaches and scaled prepared states have different stores. These examples give no universal joint limit.

Opening an energized physical winding requires its switch, arc and clamp laws [OP-TRF-02]. Perfect coupling, vanishing capacitance and other constrained limits are separate models with separate admissible preparations [OP-TRF-04].

5.4 Ideal terminal matching and finite independent stores

Two ideal transformers with common secondary nodes P, C satisfy $V_X - V_C = h_X(V_P - V_C)$ and $j_X = -h_X i_X$ for $X = A, B$. KCL gives $j_A + j_B = 0$ and $i_C = -(i_A + i_B)$, hence $(i_A, i_B, i_C) = i_A(1, -R, R - 1)$. With a dimensional scale α , the rotational mobility mapping is

$$V = \alpha\omega, \quad I = \tau/\alpha, \quad VI = \tau\omega, \quad C = J/\alpha^2. \quad (39)$$

No factor 2π is omitted if rates were originally in revolutions per second. The nine mechanical powers equal signed terminal products $(V_X - \alpha\Omega)i_X$; complete winding powers instead use $V_X - V_C$. Consequently the maximum complete-winding ratio is $\mathcal{C}_C$ and that of the three external plus four winding ports is $\max(1, \mathcal{C}_C)$. For $R = -341/100$ with B held and observation fixed to A , the mapped ratio is $341/100$

but the complete-port ratio is one. Their exact difference is $-241/100$. For $R = 2929/2950$ with A held, the winding and carrier ratios are $2950/21$ while the ground terminal ratio is one. These compare explicitly different observables; each corresponding signed work remains identical under the ideal map. Constant voltage on an ideal transformer is not a claim about indefinitely sustainable excitation of a finite core.

A dynamic circuit needs node capacitors and elastic counterparts to magnetic states. For incidence matrix B , positive diagonal C , positive definite L , and nonnegative R , independently specify the two models

$$C\dot{v} = I_{\text{ext}} - Bi, \quad Li = B^T v - Ri, \quad (40)$$

$$J\dot{\omega} = \tau_{\text{ext}} - Bf, \quad \dot{z} = B^T \omega - Mf, \quad f = Kz, \quad (41)$$

$$J = \alpha^2 C, \quad K = \alpha^2 L^{-1}, \quad M = R/\alpha^2. \quad (42)$$

Substitution of $v = \alpha\omega$, $i = f/\alpha$, $z = Li/\alpha$ proves a power-preserving correspondence for compatible initial states. Capacitor energy maps to inertia, magnetic energy to elastic energy $z^T Kz/2$, and resistor loss to $f^T Mf$. Each source, receiver, and loss is independently integrated with (2); the stores are independently evaluated. For a compound pair use $L_X = L_0 \begin{pmatrix} h_X^2 & kh_X \\ kh_X & 1 \end{pmatrix}$ and incidence columns $e_A - e_C, e_P - e_C, e_B - e_C, e_P - e_C$. A rational finite control has $L_0 = 1$, $k = 1/2$, every retained node capacitance one, each resistance one, zero initial state, a unit current source at A , and a unit conductance receiver at B over one second. Together with any specified geometry in configuration III these coefficients fully determine every closed matrix work. A rigid gear and a three-terminal winding pair do not acquire these extra degrees of freedom by a voltage rename; the dynamic correspondence is to the declared elastic model.

A physical accelerating electrical reference needs supplied compensation. For a capacitor between A and R , write $w = V_A - V_R$, $C = J/\alpha^2$, $V_R = \alpha\Omega$, and

$$C\dot{w} = I_d + I_e, \quad I_d = \tau/\alpha, \quad I_e = -C\dot{V}_R, \quad (P_d, P_e) = (wI_d, wI_e). \quad (43)$$

This realizes $w = \alpha(\omega - \Omega)$ only with compatible initial data and its two floating-source supply ports. For $C = J = \alpha = 1$, $\tau = 0$, $\Omega = 2t$, zero initial states, $w = -2t$, $I_e = -2$: $W_d = 0$, $W_e = 2$ J, $E_C(0) = 0$, $E_C(1) = 2$ J. Disabling compensation gives $W_e = 0$, $w = 0$, and $E_C(1) = 0$. A common coordinate shift on the unchanged physical capacitors also leaves their physical energy unchanged. It therefore misses the independent relative-kinetic quantity $-\Omega H + J\Omega^2/2$.

A separate reference driver $C_d = 1/200$ F, $R_d = 3/4$ Ω has $I_r = C_d \dot{V}_R = 1/100$ A and command $u_r = 2t + 3/400$ V. Its independently integrated works are $W_r = 403/40000$ J and heat $-3/40000$ J, with stores 0, $1/100$ J. Floating-source return currents cancel at R for this cell; their nonzero powers still require supplies.

For a finite actuator with current response time $\tau_a > 0$, prescribe $\tau_a \dot{j} + j = -Ca$, $C\dot{w} = j$, and actual actuator effort $e = w + R_a \dot{j} + L_a \ddot{j}$, with zero initial states and $V_R = at$. The exact solution and separate works on $[0, T]$ are

$$j = -Ca(1 - e^{-t/\tau_a}), \quad w = -a[t - \tau_a(1 - e^{-t/\tau_a})], \quad (44)$$

$$W_C = \frac{1}{2}Ca^2[T - \tau_a(1 - b)]^2, \quad Q_a = R_a C^2 a^2 [T - 2\tau_a(1 - b) + \frac{1}{2}\tau_a(1 - b^2)], \quad (45)$$

$$W_{a,s} = W_C + Q_a + \frac{1}{2}L_a C^2 a^2 (1 - b)^2, \quad b = e^{-T/\tau_a}. \quad (46)$$

The three terms in $W_{a,s} = \int ejdt$ are obtained by integrating wj , $R_a j^2$, and $L_a j \dot{j}$ separately. Actuator output work is $-W_C$, heat work $-Q_a$, and its final independent store is $L_a j(T)^2/2$. For $C = 1$, $a = 2$, $T = 1$, $\tau_a = 1/10$, $R_a = L_a = 1$ in SI units, $E_C(1) = (9 + e^{-10})^2/50$ J. Its difference from 2 J is strictly negative because $0 < \tau_a(1 - e^{-T/\tau_a}) < T$. Separate actuator and driver balances have zero exact residual even when their state correspondence fails.

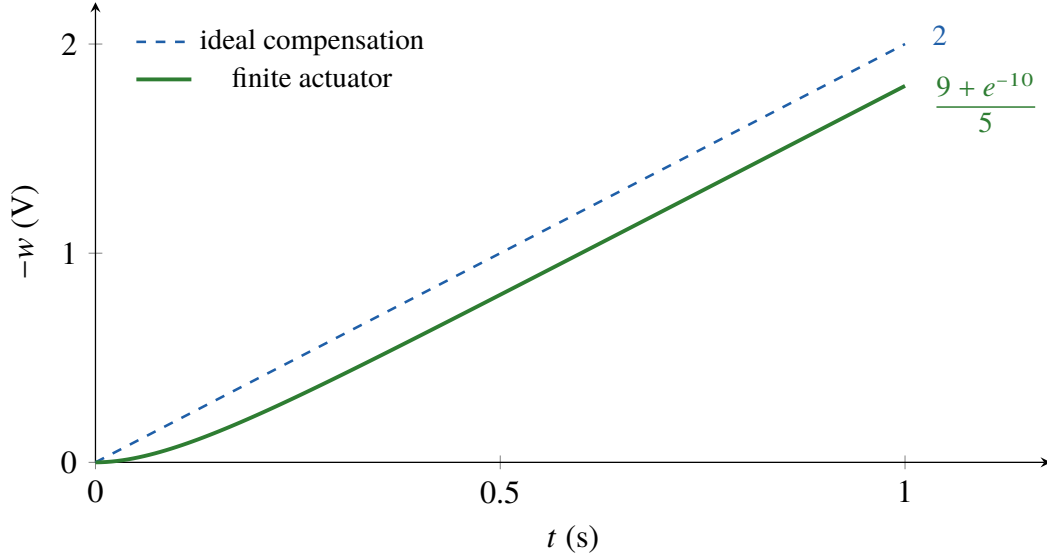


Figure 5. Exact reference-cell response from equation (46), with $C = 1$ F, $a = 2$ V/s, and $\tau_a = 1/10$ s. The plotted quantities are the stated closed forms, with their exact one-second endpoints.

Figure 5 shows the exact finite tracking departure. Finite sensors, converter losses, command limits, and a finite rail need additional constitutive states; their calibration is not supplied by (42). Existing bounded smooth-model correspondence and event results do not establish unrestricted hardware duality. A nonzero reference jump ΔV imposed in time δ through fixed $R_d, C_d > 0$ has

$$Q_d = R_d C_d^2 \int_0^\delta \dot{V}_R^2 dt \geq \frac{R_d C_d^2 (\Delta V)^2}{\delta}. \quad (47)$$

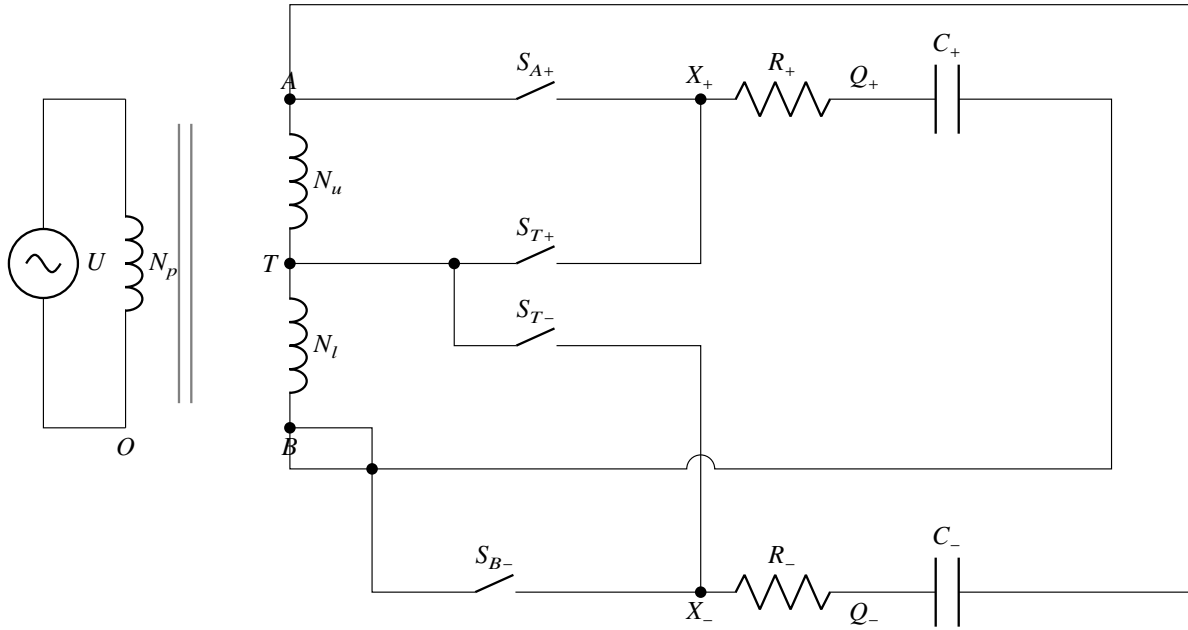
Cauchy–Schwarz proves the bound, with equality for a linear ramp. It diverges as $\delta \rightarrow 0$ for this driver; finite slow comparisons do not realize that event. With $L_0 = \epsilon^{-1}$ and $k = 1 - \epsilon^2$, the pair’s common and differential magnetic eigenvalues scale differently. A prepared common current proportional to $\sqrt{\epsilon}$ retains finite energy while the current vanishes. For example $(hi_1, i_2) = (\sqrt{\epsilon}/2, \sqrt{\epsilon}/2)$ gives $E_m = (2 - \epsilon^2)/4$ J, tending to 1/2 J. Zero common preparation gives zero instead. Thus convergence of selected port works or small differential voltage does not determine convergence of all winding states and stores. Fixed-effort clamps and unchanged passive source/load circuits are different limiting initial-value problems.

The ideal terminal map leaves physical inertial, magnetic and controller states to be identified before claiming a loaded dynamic correspondence [OP-TRF-09]. Complete physical-port ratios also require the actual return and supply ports [OP-TRF-11]. The finite-reference actuator’s separately predicted departure is a bounded model result; its physical supply and tracking accuracy remain measurable [OP-TRF-12].

6 A tapped secondary with two switched receiving branches

Murray's SERPS Figure 11 supplies a concrete switched-return architecture: an isolated primary, a secondary tap, and two resistor–capacitor branches selected on opposite waveform polarities Murray (2017). The following finite component values are declared model choices. Figure 6 names every persistent winding section and branch. The primary return O is electrically separate from secondary reference B . A selector changes an attachment; it does not discard an energized winding section.

Isolated primary, persistent secondary sections, two switched RC branches



Cell orientations: $v_+ = V_{Q_+} - V_{B_+}$, $v_- = V_{Q_-} - V_{A_-}$.

Selector edges, copper, leakage, parasitics, clamps and gate supplies follow the finite equations.

Figure 6. Schematic of the isolated primary and two-section secondary with four selectors. The positive RC branch ends at B; the negative branch ends at A. Each switch is a finite conductance in the model, with separately supplied gate state. Winding polarities, leakage and mutual coupling are specified by the inductance matrix. The dashed magnetic connection is not a conductor between primary and secondary.

The four switches connect $A - X_+$, $T - X_+$, $T - X_-$ and $B - X_-$. Receiver resistors join X_+ to Q_+ and X_- to Q_- . The positive cell is $Q_+ - B$ and the negative cell is $Q_- - A$. For a one-second sinusoidal drive, the nominal commands are:

Table 7. The connected state and capacitor voltage determine actual current direction; the clock alone does not establish it.

Quarter	Selected connection	Intended branch operation
Positive rising	$A - X_+$	Charge through full secondary
Positive falling	$T - X_+$	Return through lower section

Quarter	Selected connection	Intended branch operation
Negative increasing magnitude	$B - X_-$	Charge through full secondary with opposite orientation
Negative falling magnitude	$T - X_-$	Return through upper section

For an ideal positive charging interval, $v_s = Ri + v_C$ and $C\dot{v}_C = i$. On return, $v_C = Rj + fv_s$ and $C\dot{v}_C = -j$, where j enters the winding and f is the tap fraction. The condition $fv_s < v_C < v_s$ permits both directions under the respective connections. For $R = C = 1$ in SI units, a constant connected voltage v_s gives $v_C = v_s + (v_0 - v_s)e^{-t}$ on $[0, T]$. With $a_0 = v_0 - v_s$, separate integration gives

$$W_s = -v_s a_0 (1 - e^{-T}), \quad Q_R = \frac{a_0^2}{2} (1 - e^{-2T}), \quad (48)$$

$$\Delta E_C = \frac{1}{2} \{ [v_s + a_0 e^{-T}]^2 - v_0^2 \}. \quad (49)$$

Units are joules when the displayed constants use volts, ohms, farads and seconds. Substitution verifies $W_s - Q_R = \Delta E_C$. The negative branch has its own orientation and prepared voltage. $v_0 = v_s$ gives zero current; zero preparation can instead give nonzero charging current. Finite leakage prevents assigning an instantaneous winding current reversal to a physical tap event.

6.1 Persistent states and finite commutation

Let $v = (V_A, V_T, V_{X+}, V_{Q+}, V_{X-}, V_{Q-})$ relative to B . Use three winding currents in primary, upper-secondary and lower-secondary order, and incidence columns $B_w = (0, e_A - e_T, e_T)$. With turns vector $N = (1, \sigma(1 - f), \sigma f)$, define

$$L = (1 - \kappa) \text{diag}(N_f^2) + \kappa N N^T, \quad 0 < f < 1, \quad 0 \leq \kappa < 1, \quad (50)$$

with unit henry scale. The tap fraction f and magnetic coupling κ are independent. Positive leakage makes L positive definite. For $f = \kappa = 1/2$, $\sigma = 1$, $L = \begin{pmatrix} 1 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/8 \\ 1/4 & 1/8 & 1/4 \end{pmatrix}$ H. Each receiving resistor is 1Ω , each cell $1/10$ F, and each node has parasitic $1/50$ F to B :

$$C_n = \frac{1}{50} I + \frac{1}{10} e_{Q+} e_{Q+}^T + \frac{1}{10} (e_{Q-} - e_A)(e_{Q-} - e_A)^T. \quad (51)$$

All selector paths have $R_{\text{on}} = 1/20 \Omega$ and $R_{\text{off}} = 10000 \Omega$. With source resistance $1/5 \Omega$ and each winding resistance $1/20 \Omega$, the independent circuit equations are

$$Li = B_w^T v + e_p U - R_t i, \quad C_n \dot{v} = -B_w i - G(t)v - i_{\text{cl}}, \quad R_t = \text{diag}(1/4, 1/20, 1/20). \quad (52)$$

Every resistor or switch with incidence a_j contributes $g_j a_j a_j^T$ to G . Clamps across $A - B$, $T - B$, and each cell use $i_{\text{cl},j} = 5 \text{sgn}(w_j) \max(|w_j| - 2, 0)$ in amperes. Each gate is a $1/10000$ F capacitor supplied through 20Ω . Prescribed selector commands and the gate circuit are separate model laws; actual semiconductor coupling and actuator cost need identification.

For finite edges use $s(\xi) = 3\xi^2 - 2\xi^3$ on $0 \leq \xi \leq 1$ to interpolate conductance between its off and on values. Specify opening before closing for a gap, or closing before opening for overlap. Each path

has its own width and edge endpoints. One initialized commutation uses $i = (1/10, 1/5, -1/10)$ A and $V_{Q+} = V_{Q-} = 3$ V, with other nodes zero; the clamp law therefore participates. The separate zero-state history drives two one-second periods with $U = \sin(2\pi t)$ V, then sets source and gates to zero and connects $1/5$ S node bleeders for one second. Its actual final states remain. Finite positive L, C_n give continuous states and zero impulse at these events, even when a commanded coefficient or power jumps.

The reactive store is $(i^T L i + v^T C_n v)/2$ plus all gate stores. For a boundary including thermal matter H , source $U i_p$ and gate-supply products enter; receiver powers $-(V_{X+} - V_{Q+})^2$ and $-(V_{X-} - V_{Q-})^2$ and bath $-H/2$ leave. Copper, source resistor, selector, clamp and gate-resistor products enter $\dot{H}$ as internal conversion. An electrical-only boundary instead exports those losses and excludes H . Both endpoints are evaluated independently under the chosen convention.

On a fixed linear interval the exact primitive (2) computes each work independently. A smooth edge uses its fundamental matrix $\Phi(t, t_0)$, defined by $\dot{\Phi} = A(t)\Phi$, $\Phi(t_0, t_0) = I$, with the forcing convolution. The separate integral $\int y_0^T \Phi^T Q_j(t) \Phi y_0 dt$ defines that edge's work. Where a clamp changes branch, its guard and event sides must also be retained. This conditional exact formulation preserves the finite-event question without assigning a numerical trajectory or a unique zero-width limit. Removing leakage or parasitics imposes additional constraints and can exclude the chosen initial state.

6.2 A parallel/series bank is another connection graph

The patent's Figure 19 specifies a parallel/series network block without resolving its individual cell selectors Murray (2017). A declared finite realization uses cells $X_1 - Y_1$ and $X_2 - Y_2$, charge selector $A - X_1$, injection selector $T - X_1$, parallel links $X_1 - X_2$ and $Y_1 - Y_2$, and series link $Y_1 - X_2$. Its 1Ω receiver joins $Y_2 - B$. This receiver attachment differs from the later hidden-state experiment's $X_1 - Y_2$ attachment.

Take $C_1 = 1/10$ F, $C_2 = 1/10$ or $3/20$ F, with the same node parasitics, finite selector resistances, windings, clamps and gate laws as above. Command charge and both parallel links during the first and third quarters, injection and series during the second and fourth. The series-to-parallel return at half a second retains the actual cell voltages. Zero preparation and a separately initialized unequal-cell case with $(v_1, v_2) = (1, 1/2)$ V require their own source, switch, receiver and endpoint expressions. Those expressions follow from the new incidence matrices and finite-edge maps; the initialized case supplies no preparation work by itself.

An isolated pair gives a useful exact control. For $(C_1, C_2) = (1, 2)$ F, initial $(v_1, v_2) = (1, 0)$ V and exchange resistance 1Ω , $v_1 = 1/3 + 2e^{-3t/2}/3$ and $v_2 = 1/3 - e^{-3t/2}/3$. On $[0, 1]$ s, the independent heat integral is $(1 - e^{-3})/3$ J and the capacitor endpoints are $1/2$ and $1/6 + e^{-3}/3$ J. Positive resistance gives continuous voltages without an impulse. Its isolated single-path zero-resistance heat limit is unique; several competing paths need their own partition laws. This result does not determine the transformer-connected bank's work.

6.3 Clocked cycles, magnetic limits, and physical service

For the linear fixed-clock graph, compose interval maps to obtain $x(T) = Mx(0) + b$. A periodic state satisfies $(I - M)x_* = b$. Existence, uniqueness and attraction are different tests: respectively $b \in \text{range}(I - M)$, invertibility of $I - M$, and $\rho(M) < 1$. With $M = I, b = 0$ every state is fixed; $M = I, b \neq 0$ has none; $M = 2I$ has a unique unstable fixed state. The remaining physical question is whether a declared clocked realization stays within magnetic and controller limits [OP-TRF-13].

Theorem 2 (Clocked contraction). *For the above network with inactive clamps, fixed positive-definite L, C_n , strictly positive winding resistance and uniformly grounded positive conductance, each periodic*

forcing and clock has one attracting periodic electrical state.

Proof. The difference of two forced solutions obeys the homogeneous laws. Multiplication by its own currents and voltages gives $\dot{E} = -i^T R_i i - v^T G(t) v$. The declared positive resistances and connected off paths imply $\text{diag}(R_i, G(t)) \geq \eta \text{diag}(L, C_n)$ for some $\eta > 0$ on the compact clock period. Hence $\dot{E} \leq -2\eta E$ and the period map contracts its energy norm by at most $e^{-\eta T} < 1$. The affine map has one attracting fixed point. The same estimate covers continuous finite edges. $\square$

Gate RC states have their own affine fixed points. A thermal law $\dot{H} = D(t) - H/2$ has the period relation $H(T) = e^{-T/2} H(0) + \int_0^T e^{-(T-t)/2} D(t) dt$. An insulated thermal state with positive conversion does not repeat merely because voltages do. Repeated flux endpoints also leave continuous peak flux and material admissibility to be checked.

A state-triggered controller needs its own map. One example switches at $V_{Q+} = 1/100$ V upward and $1/200$ V downward. For a guard with normal n , continuous state and vector fields f^-, f^+ , its transverse-event variation contains $S = I + (f^+ - f^-)n^T / (n^T f^-)$. This formula requires $n^T f^- \neq 0$; grazing, simultaneous guards and changed event order require separate treatment. A clocked contraction theorem does not supply full-wave hybrid attraction.

Two material extensions illustrate the needed constitutive information. With positive leakage D_L and inductance scale $L_m = 1$ H, let $\lambda = D_L i + L_m \kappa I_s \tanh(N^T i / I_s) N$. Integrating $i^T d\lambda$ gives

$$E_m = \frac{1}{2} i^T D_L i + L_m \kappa [I_s q \tanh(q/I_s) - I_s^2 \log \cosh(q/I_s)], \quad q = N^T i. \quad (53)$$

Here q is a current coordinate. At $I_s = 1/2$ A, a 4 V sinusoidal drive defines a finite amplitude comparison against the fresh linear model. This reversible saturating law has no remanence or hysteresis state. A separate temperature law $R_w = (1/20)(1 + H/(100 \text{ J})) \Omega$, for the stated thermal scaling, changes the trajectory through heating. Neither extension inherits the constant-matrix theorem without a new proof. Compare complete service only after fixing its definition: for example total receiver heat on $[0, 3]$ s, both zero-state preparations, source $\sin(2\pi t)$ on $[0, 2]$ and zero afterward, and the actual finite bleeder reset. A full-secondary baseline retains both secondary sections and selects its load on a declared descending branch of the service equation before cost comparison. If several branches exist, their selection is part of the contract. Retain each endpoint, even if the selected service is equal. Gross source draw and return, net source work, gate work and useful delivery are separate integrals. A source model supplies one finite ordering; an exact general efficiency ordering is not established by the formulation here. Matched reset states or their explicit costs are additional requirements for a closed-cycle comparison.

6.4 The generator boundary

Enclosing the generator makes capacitor-to-generator electrical return internal. Prime-mover shaft work, field excitation and controller supplies remain external, with rotor, magnetic, field and thermal stores at both ends. The physical prime-mover comparison requires the actual flux/coenergy law and independently established motion and excitation [OP-TRF-14].

A bounded ideal converter control shows the signs. Choose $\omega = 2\pi$ rad/s, $e = \sin(2\pi t)$ V and $i = \pm \sin(2\pi t) + \frac{1}{2} \cos(2\pi t)$ A on $[0, 2]$ s, with converter torque ei/ω . Let rotor inertia be $1/100$ kg m², drag $D = 1/1000$ N m s, and a separate unit-inductance unit-resistance field carry 1 A under 1 V. Separate products give electrical delivery ± 1 J, shaft input $\pm 1 + \pi^2/125$ J, field input 2 J, and drag/field heat exports $-\pi^2/125, -2$ J. Rotor and field stores remain $\pi^2/50$ and $1/2$ J at both independently evaluated ends. The imposed reciprocal converter and decoupled field are explicit assumptions. Their positive or negative shaft

input does not determine a physical machine's complete-cycle ordering.

Finite disconnected preparation is also explicit. A unit RL field driven by constant voltage U from zero has $i = U(1 - e^{-t})$, $W_s = U^2[T - 1 + e^{-T}]$, resistor heat $U^2[T - 2(1 - e^{-T}) + (1 - e^{-2T})/2]$, and store $U^2(1 - e^{-T})^2/2$. Setting its source to zero through the resistor gives $i = i_0 e^{-t}$, heat $i_0^2(1 - e^{-2T})/2$ and a nonzero finite endpoint. A rotor accelerated by constant torque $J\omega_*/T$ with zero drag receives $J\omega_*^2/2$ by its own torque–rate integral. These separate paths clarify what remains to be specified when actual field and armature dynamics are coupled.

6.5 State and observation correspondence

For the fixed linear charging topology, the selected outputs ($i_p, V_A, V_T, V_{Q+}, V_{Q-} - V_A$) and known component laws give an observability matrix $[H; HA; \dots; HA^8]$ of rank nine for the nine electrical states. This can be seen directly with inactive clamps. The measured last output and V_A give V_{Q-} . The two cell-node laws give

$$V_{X+} = V_{Q+} + \frac{3}{25} \dot{V}_{Q+}, \quad V_{X-} = V_{Q-} + \frac{3}{25} \dot{V}_{Q-} - \frac{1}{10} \dot{V}_A, \quad (54)$$

with the coefficients in seconds for the stated resistor and capacitance values. All six node voltages are now known. Put $r = C_n \dot{v} + Gv$; the A, T node laws then give $i_u = -r_A$ and $i_l = i_u - r_T$. These two components need only the already observed node derivatives. Together with measured i_p , they recover all nine states, proving the rank statement. This is structural reconstruction from ideal traces. Temperature-independent elements leave $H_{\text{thermal}}(0) = 0$ or 1 J electrically indistinguishable while their thermal difference at time T is $e^{-T/2}$ J. A fixed-topology rank test gives neither a noise bound nor a switched observation certificate. Individual cell and material observations still belong to the full endpoint account. Even the nonnegative cell pairs (1, 1) and (0, 2) V share a 2 V series terminal with unit-cell stores of 1 and 2 J.

The analogous mechanical ambiguity is already present in unit-inertia states $(\omega_c, \omega_p) = (0, 1)$ and (10, 11) rad/s: their relative rate is one, but their kinetic stores are 1/2 and 221/2 J. Absolute unwrapped motion and signed effort must be observed together. An encoder count is $2\pi/N$ radians for N counts per turn; pulse totals without reversal direction lose signed work. For a 1 F rail at 1000 V, a positive 1/1000 V endpoint error changes the inferred energy by 2000001/2000000 J, from the exact expression $Cv\delta v + C(\delta v)^2/2$. Small relative voltage error therefore does not guarantee a small absolute energy error. A full low-energy rig target of 1/100 J needs independently bounded channels, endpoint covariance, event timing and bandwidth; it is not established by naming the instruments.

A finite mechanical counterpart can be constructed with $J = \alpha^2 C_n$, $z = \lambda/\alpha$, $K = \alpha^2 L^{-1}$ and $i = Kz/\alpha$. Six rotors supply node inertias; two differential flywheels supply the capacitor terms between nodes. Three positive modal springs realize K . At $\alpha = 1$, $K = \begin{pmatrix} 3/2 & -1 & -1 \\ -1 & 6 & -2 \\ -1 & -2 & 6 \end{pmatrix}$ and $LK = I$. Its independent equations are $J\dot{\omega} = -B_w Kz - G\omega - \tau_{cl}$ and $\dot{z} = B_w^T \omega - R_t Kz + b_{\text{anchor}}$. An anchored drive supplies the primary forcing; every dashpot, gate rotor and thermal port must map as well. Substitution proves state, power and constitutive-store correspondence for the declared laws. The rigid planetary train has two independent speeds, whereas this circuit has six independent capacitive node coordinates. A bijection with that unchanged rigid train is impossible. The added rotors, differentials, springs and actuators are distinct physical hardware.

7 Signed states and a reconnection experiment

7.1 Four quantities that an instrument must distinguish

An oriented charge, its absolute value, its energy and its work have different units and different laws. Fix each capacitor's physical plate orientation before defining $q_j = C_j v_j$. Reversing a sensor changes its reported sign; reconnecting plates changes the incidence matrix and can change the conserved quantities. On a fixed isolated two-cell exchange, KCL gives $\dot{q}_1 = -i$, $\dot{q}_2 = i$, so $Q = q_1 + q_2$ is constant. For $C_* = C_1 C_2 / (C_1 + C_2)$ and $d = v_1 - v_2$,

$$E_C = \frac{Q^2}{2(C_1 + C_2)} + \frac{1}{2} C_* d^2, \quad A_q = |q_1| + |q_2|. \quad (55)$$

Neither expression is a signed port work. With a resistor R , $d = d_0 e^{-t/(RC_*)}$ and its separately integrated heat is $C_* d_0^2 (1 - e^{-2T/(RC_*)})/2$. The absolute charge sum is nonincreasing: when the signs agree its derivative is zero; when they oppose, $\dot{A}_q = -2|i|$ until a charge reaches zero. This sign argument depends on the resistor law and the declared orientations.

An inductor permits a different exact result. With initial charges $(Q, 0)$ and zero current,

$$q_1 = \frac{Q(C_1 + C_2 \cos \omega t)}{C_1 + C_2}, \quad q_2 = \frac{QC_2(1 - \cos \omega t)}{C_1 + C_2}, \quad i = \frac{QC_2 \omega \sin \omega t}{C_1 + C_2}, \quad \omega^2 = \frac{1}{LC_*}. \quad (56)$$

For $(C_1, C_2, L) = (1, 19, 20/19)$ in SI units and $Q = 10$ C, the states at 0 and π s are $(10, 0, 0)$ and $(-9, 19, 0)$. Thus $Q = 10$ C, A_q grows from 10 to 28 C, and total reactive energy is 50 J at both ends. Integration of $-v_1 i$, $v_2 i$ and $Li\dot{i}$ separately gives $-19/2$, $19/2$ and 0 J. No external work enters this isolated boundary. The inductor's nonzero intermediate store is essential.

A driven lossy winding has no corresponding universal winding-only linear invariant. If $a^T i$ were conserved for all inputs and states in $Li\dot{i} = bU - Ri$, it would require both $a^T L^{-1}b = 0$ and $a^T L^{-1}R = 0$. Invertible positive L, R force $a = 0$. At every topology change the applicable invariant must be derived anew.

7.2 The same terminal history can hide four joules

Connect two equal cells in series through a receiver R_L . For cell voltages v_1, v_2 , set $s = v_1 + v_2$ and $d = v_1 - v_2$. The common current gives

$$\dot{s} = -\frac{2s}{R_L C}, \quad \dot{d} = 0, \quad E_C = \frac{C}{4}(s^2 + d^2), \quad W_L = -\frac{Cs_0^2}{4}(1 - e^{-4T/(R_L C)}). \quad (57)$$

With $C = 1$ F, preparations $(1, 1)$ and $(3, -1)$ V both give $s_0 = 2$ V and the same entire terminal history. Their independently evaluated initial stores are 1 and 5 J; the unobserved difference $d = 0$ or 4 V retains a 4 J energy separation at every time. Preparation II contains a reversed cell.

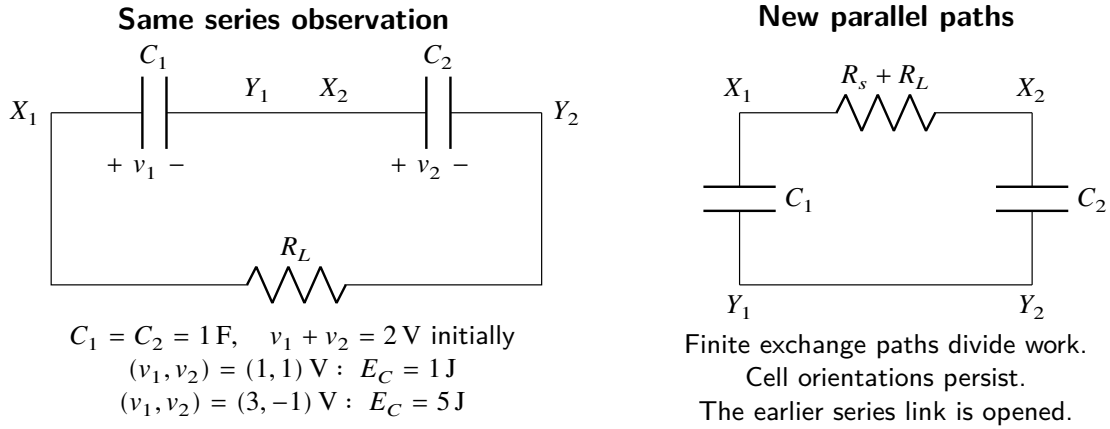


Figure 7. Two equal series cells can have the same terminal voltage and different internal energy. A later parallel connection exposes the difference mode only through its declared finite paths; the individual cell orientations remain fixed.

Reconnecting corresponding plates through a finite resistance can expose that difference mode. If a receiver and selector resistance are in series, the receiver receives the fraction $R_L / (R_L + R_s)$ of the converted difference energy, with finite-time factor $1 - e^{-2T / ((R_L + R_s)C_*)}$. The selector receives the complementary fraction. Residual cell energy remains at finite T . This calculation specifies neither the preparation work nor the useful delivery in a different switched graph.

7.3 A finite switched observation map

The bounded reconnection result concerns identification of two known preparations by a loaded cell probe [OP-TRF-15]. Its exact finite formulation below states the component laws and separation inequalities required for that binary certificate. Recovering arbitrary cell signs, energies or temperatures is a different inverse problem.

Use nodes $(A, T, X_1, Y_1, X_2, Y_2)$ relative to B , unit-farad cells across $X_1 - Y_1$ and $X_2 - Y_2$, and $1/50 \text{ F}$ parasitics at every node. A 1Ω receiver joins $X_1 - Y_2$. Selectors join $A - X_1, T - X_1, X_1 - X_2, Y_1 - Y_2, Y_1 - X_2$ and the probe path $X_1 - Y_1$. Use the winding matrix (50) with $f = \kappa = 1/2$, each winding resistance $1/20 \Omega$, source resistance $1/5 \Omega$, and zero source voltage during operation. The five ordinary selectors have on/off resistances $1/20$ and 10000Ω ; the probe has 10 and $10^6 \Omega$. Cell and terminal clamps have slope 5 S above 4 V magnitude. Gate capacitance and resistance remain $1/10000 \text{ F}$ and 20Ω .

The initial node vectors are

$$v_0^I = (0, 0, 2, 1, 1, 0), \quad v_0^{II} = (0, 0, 2, -1, -1, 0) \quad \text{V.} \quad (58)$$

Both terminal voltages $V_{X_1} - V_{Y_2}$ are 2 V ; both parasitic stores are $3/50 \text{ J}$, independently of the 1 or 5 J cell store. Initial winding currents are zero. The series selector is on until $t = 1/4 \text{ s}$. During the next $1/100 \text{ s}$, smooth conductance edges open it and close both parallel selectors and the probe. At $t = 1 \text{ s}$, source and gates are turned off and $1/5 \text{ S}$ node bleeders operate until $t = 2 \text{ s}$. All final states remain in the account. Finite off conductance means that the two earlier terminal histories need not agree exactly as they do in (57).

Let x collect winding and node states, and append the probe filter $\dot{z} = (V_{X_1} - V_{Y_1} - z)/(1/50 \text{ s})$ with

zero initial filter state. On each prescribed clamp branch the augmented linear or affine equations have a fundamental matrix. Their ordered composition defines the switched observation $z(t) = \mathcal{O}_t(x_0)$. It includes the probe conductance in the physical state equations; filtering is not a license to omit probe work.

For an exact conditional certificate, calculate $z_j = \mathcal{O}_{7/20}(x_0^j)$ and form the two uncertainty intervals $I_j = [z_j - u_z, z_j + u_z]$, $u_z = 1/20$ V. If $|z_{II} - z_I| > 2u_z$ and the precommutation terminal difference is bounded by $1/10$ V throughout its window, these readings distinguish the two specified preparations. Since the two candidate cell energies are known exactly, correct binary classification meets a $1/100$ J energy target in the exact-coefficient model. Those inequalities, including continuous-time preterminal bounds, are the certificate conditions for the finite model; no sampled voltage is an all-state observability proof. Physical acquisition must bound component drift, probe loading, gain, bandwidth, polarity, timing and preparation uncertainty before adopting these decision intervals. Unknown analog preparations require a new uncertainty analysis rather than a two-label classifier.

7.4 Paired work includes the chargers and the retained states

The same receiver and clock define a bounded paired-work comparison [OP-TRF-16]. More prepared energy alone fixes no sign for the difference of that receiver's work. The work is determined by the loaded trajectory and every connected conversion path.

Preparation is physical. Disconnect windings and selectors and charge the six-node capacitance matrix C_b from zero during $[-1, 0]$ s using six independent 1Ω Thevenin sources. For constant source vector u , $C_b \dot{v} = u - v$, whence $v(0) = (I - e^{-C_b^{-1}})u$. The positive matrix C_b makes the charge map invertible, so each vector in (58) has its own specified finite u . Each source work is $\int u_j(u_j - v_j)dt$; each resistor exports $-\int (u_j - v_j)^2 dt$. The endpoint is independently $v(0)^T C_b v(0)/2$.

During operation evaluate separately, for each preparation,

$$W_L^j = - \int_0^2 (V_{X_1}^j - V_{Y_2}^j)^2 dt, \quad W_{\text{probe}}^j = - \int_0^2 g_{\text{probe}}(t) (V_{X_1}^j - V_{Y_1}^j)^2 dt. \quad (59)$$

Source, selector, clamp, gate and bath products have their own integrals. With thermal matter retained, all internal resistive conversions enter $\dot{H} = D - H/2$; the bath work is $-\int H/2 dt$. The total endpoint includes winding, cell, parasitic, gate and thermal stores. For each event retain the same continuous state on both sides and both one-sided powers. Finite positive parameters give zero impulse.

On linear intervals the independent quadratic primitive (2) evaluates these works; finite edges use their ordered fundamental matrix and separate quadratic integrals. The sign of $W_L^{II} - W_L^I$ follows from those expressions, not from the 4 J initial cell difference. A physical comparison requires an uncertainty below its predicted separation and retains actual reset endpoints. It establishes neither equal useful service nor a closed-cycle generator ranking.

7.5 An all-time sign and magnitude certificate

A finite topology change can be treated without inferring sign events from sampled states. Retain the LC path in (56) with $(C_1, C_2, L) = (1, 19, 20/19)$ and add a 1Ω shunt across cell 1 only on $[a, b] = [\pi/2, \pi/2 + 1/10]$ s. On the full window $[0, 2]$ s, write its conductance as $g = 0$ or 1 S. The equations and independent energy are

$$\dot{q}_1 = -i - gq_1, \quad \dot{q}_2 = i, \quad \dot{i} = \frac{19q_1 - q_2}{20}, \quad E = \frac{q_1^2}{2} + \frac{q_2^2}{38} + \frac{10i^2}{19}. \quad (60)$$

Here $\dot{E} = -gq_1^2$ and $\dot{Q} = -gq_1$. The old conserved charge is therefore not conserved during the shunt. For initial state $(10, 0, 0)$, Cauchy–Schwarz gives, at every time, $A_q^2 \leq 40E_C \leq 40E \leq 2000C^2$. The identically zero initial state remains zero. This is a conservative bound for these preparations and components, not a universal multiplier.

Before a , (56) gives $(q_1, q_2, i)(a) = (1/2, 19/2, 19/2)$ in SI units. $q_2 = 0$ has its initial tangency at $t = 0$; both charges are positive on $(0, a]$. Energy bounds give $|i| < 10$ and $|q_2| < 44$. On the shunt interval, putting $\tau = t - a$, the integrating-factor solution of $\dot{q}_1 + q_1 = -i$ gives $|q_1| \leq 1/2 + 10\tau \leq 3/2$. Also $17/2 \leq q_2 \leq 21/2$ and $|i| \leq 39/20$, hence $i \geq 1861/200 > 9$. Thus $\dot{q}_1 < 0$ throughout this interval. While $q_1 \geq 0$, it satisfies $-21/2 < \dot{q}_1 \leq -9$. Consequently its unique zero obeys

$$\frac{\pi}{2} + \frac{1}{21} < t_* < \frac{\pi}{2} + \frac{1}{16}. \quad (61)$$

Before that zero $\dot{A}_q = -q_1 < 0$; after it, $\dot{A}_q = 2i + q_1 > 0$. The zero is a magnitude cusp with distinct one-sided derivatives, not a differentiable minimum.

After b , the energy bound also gives $|q_1| \leq 10$ and $|i| < 117/10$. Since $2 - b < 2/5$, $i > 1861/200 - 117/25 = 37/8 > 0$ to the final time. Therefore q_1 stays negative, q_2 stays positive, and there are no further charge zeros. At a, b states are continuous; the appropriate one-sided vector fields give the switched derivatives. This covers every time, both preparations, the tangency, persistent zeros, single crossing, cusp and both switch sides.

The independent shunt work is $W_h = -\int_a^b q_1^2 dt$. On each constant- g interval set $x = (q_1, q_2, i)^T$ and $x(t) = e^{A_g(t-t_0)}x(t_0)$ from (60); the quadratic primitive evaluates W_h separately from the two endpoint energies. Multiplying the three independent equations verifies their equality $E(2) - E(0) = W_h$. If the resistor is enclosed, its heat store replaces that export and total retained energy is constant. The finite shunt interval does not specify an actual switch's ignition, parasitic or sensor law.

8 Other prepared states and switching paths

8.1 A passive two-store circuit has several different limits

Configuration V first uses a parallel R_C, C section in series with a parallel R_L, L section. Its terminal impedance is

$$Z(s) = \frac{R_C}{1 + sR_C C} + \frac{sLR_L}{R_L + sL}. \quad (62)$$

KCL in each section and KVL in their series connection derive this formula. At $s = 1 \text{ s}^{-1}$ with untargeted components equal to one in SI units, $C = \epsilon$, $R_C = 1$ gives $Z = 1/(1 + \epsilon) + 1/2$; $C = R_C = \epsilon$ in their respective units gives $Z = \epsilon/(1 + \epsilon^2) + 1/2$. At $\epsilon = 1/10$ these are $31/22$ and $121/202 \Omega$; the limits are $3/2$ and $1/2 \Omega$. This Laplace-domain comparison is a terminal-law statement, not a work measurement or a physical component with infinite or zero value.

The exact zero/infinite topologies have distinct hypotheses. A zero capacitor removes its capacitive branch, retaining R_C ; infinite capacitance constrains its voltage to a constant only for bounded current and a declared prepared voltage. A zero inductor enforces zero branch voltage, while infinite inductance constrains current for bounded voltage. Zero resistance shorts its section; infinite resistance removes that resistor. Incompatible prepared charge or flux needs a physical regularization, not substitution into (62).

Close the terminal by a zero-voltage source, retaining finite positive components. For capacitor voltage v and inductor current i , KCL gives

$$C\dot{v} = -(R_C^{-1} + R_L^{-1})v + i, \quad L\dot{i} = -v, \quad E = \frac{1}{2}Cv^2 + \frac{1}{2}Li^2. \quad (63)$$

The complete store boundary excludes the two thermal reservoirs and the terminal source. Its powers are 0, $-v^2/R_C$, and $-v^2/R_L$. For $C = L = R_C = R_L = 1$, prepared $(v, i) = (1, 0)$ gives $v = (1 - t)e^{-t}$, $i = -te^{-t}$. On $[0, 1]$ s,

$$W_C^h = W_L^h = -\frac{1 - e^{-2}}{4} \text{ J}, \quad E(0) = \frac{1}{2} \text{ J}, \quad E(1) = \frac{e^{-2}}{2} \text{ J}. \quad (64)$$

The sum of absolute physical heat works is $(1 - e^{-2})/2$ J; the zero-state control gives zero for every work and store. This prepared experiment retains the original two-section topology. Other nondimensional coscalings require their own positive component paths, compatible endpoints, and individually resolved source and heat works.

Two explicitly different clamp regularizations illustrate why zero-state transfer functions do not fix prepared switching work. A unit capacitor at 1 V discharged through resistance $R = \epsilon$ for $T = \epsilon \ln 2$ exports $3/8$ J and retains $1/8$ J. A lossless auxiliary inductor $L = \epsilon^2$ attached instead gives $v = \cos(t/\epsilon)$, $i = \sin(t/\epsilon)/\epsilon$. At $T = \pi\epsilon/2$ capacitor energy is zero, inductor energy $1/2$ J, and each capacitor/inductor interface work is respectively $-1/2, +1/2$ J; external heat work is zero. Both follow exact finite equations and the two endpoint laws. Neither is the unique instantaneous fate of the initial charge. The finite component and time choices, their peak currents, and all additional switch or clamp paths remain part of the experimental claim.

These finite paths do not select a unique prepared singular limit [OP-LR21-01]. Smooth zero-state forcing also leaves the work of an actual prepared-state switch unspecified [OP-LR21-02].

8.2 Hidden prepared energy and its observability

A transformer-coupled series R, C branch in parallel with G_0 has

$$Y(s) = G_0 + \frac{n^2 s C}{1 + s R C}, \quad v_C(t) = V_0 e^{-t/(RC)}, \quad I_{\text{terminal}}(t) = -\frac{n V_0}{R} e^{-t/(RC)} \quad (65)$$

on a zero-voltage terminal preparation release, with the indicated current positive into the passive input. The internal resistor current has magnitude $|V_0|e^{-t/(RC)}/R$. At $n = 0$ the transformer is interpreted as the decoupled boundary topology, not an invertible zero-ratio transformer. There is no terminal hidden-mode current, but $E_C = CV_0^2 e^{-2t/(RC)}/2$.

Choose $C = 1/1000000$ F, $R = 100 \Omega$, $V_0 = 1$ V, $T = RC \ln 2 = \ln 2/10000$ s. The source work is zero, initial store $1/2000000$ J, final store $1/8000000$ J, and heat work $-3/8000000$ J by direct integration of $-v_C^2/R$. An empty mode predicts zero internal current and zero heat; a prepared hidden mode predicts initial internal current $1/100$ A and the stated heat. A branch voltage/current observation resolves the difference even when the terminal observation cannot. Calorimetric resolution at that energy would need its own demonstrated uncertainty; branch electrical measurement avoids presuming it.

For a known single nonzero n , a certified terminal current bound u_I at release implies

$$|V_0| \leq R u_I / |n|, \quad E_C(0) \leq \frac{C R^2 u_I^2}{2 n^2}. \quad (66)$$

For $n = 1/100$, $u_I = 1/1000000$ A, and the preceding R, C , this upper bound is $1/20000000000$ J. It grows without bound as sensitivity n approaches zero. Known gain, loading, and bandwidth are hypotheses of the bound. A noisy or nonideal transformer requires their independently bounded transfer law and additional stores.

Multiple modes introduce a further obstruction: two equal R, C, n branches prepared at $+V, -V$ have cancelling terminal release currents but total energy $CV^2 > 0$. With the preceding C, R, n and $V = 1$ V this is $1/1000000$ J, exceeding the single-mode bound despite exact terminal cancellation. All their initial opposite branch currents are observable separately. More generally, if boundary observations are $y(t) = He^{At}x_0$, an energy bound from them exists on a preparation subspace only if no nonzero energy-bearing direction lies in its observation nullspace, with a quantitatively bounded inverse on the retained subspace. Internal branch measurements remove this nullspace in the two-mode example. More modes, nearly coincident time constants, nonideal couplings, and correlated noise require their own finite observability and error bounds; a single terminal noise threshold cannot establish them.

Near-cancelled or unobserved modes require branch-state observations with their own uncertainty [OP-LR23-01]. Exact decoupling leaves prepared hidden energy and any later reconnection as separate physical questions [OP-LR23-02].

8.3 Common current is a physical preparation

Take two lossless inductors $L_1 = 1$ H, $L_2 = 19$ H connected through a capacitor $C = 20/19$ F. The oriented equations are $L_1 \dot{i}_1 = -v$, $L_2 \dot{i}_2 = v$, $C\dot{v} = i_1 - i_2$. For initial $(i_1, i_2, v) = (10 + u, u, 0)$, define $c = i_1 + 19i_2 = 10 + 20u$ and $d = i_1 - i_2 = 10 \cos t$. Then

$$i_1 = \frac{c + 19d}{20}, \quad i_2 = \frac{c - d}{20}, \quad v = \frac{19}{2} \sin t, \quad E = \frac{c^2}{40} + \frac{19d^2}{40} + \frac{Cv^2}{2} = 50 + 10u + 10u^2 \text{ J.} \quad (67)$$

The equality follows by substitution of the state, not an assumed constant store. The operating boundary has no external ports; the capacitor and two coil works integrate separately as changes of their explicit quadratics. On $[0, \pi]$ s, $(W_1, W_2, W_C) = (-19c/20, +19c/20, 0)$ J, so their sum is zero. For $u = 0$, this pair is $(-19/2, +19/2)$ J; for $u = -1/2$ it is $(0, 0)$; for $u = -1$ it is $(+19/2, -19/2)$.

For actual preparation, disconnect the operating capacitor and independently ramp both coil currents from zero as $i_1 = (10 + u)t$, $i_2 = ut$ on $[0, 1]$ s. With ideal lossless windings the two imposed source voltages are $10 + u$ and $19u$ V. Their independently integrated works are $(10 + u)^2/2$ and $19u^2/2$ J. Add winding resistance R_j and each source work additionally contains $R_j I_{j*}^2/3$, with separate negative heat work of that magnitude. A reversed controlled ramp supplies the explicit reset; passive relaxation is another path.

Over $-1 \leq u \leq 1/2$, completing the square gives $E = 95/2 + 10(u + 1/2)^2$ J, hence the stated range $[95/2, 115/2]$ J. At $u = 0$ initial fluxes are $(10, 0)$ Wb and final fluxes $(-9, 19)$ Wb; at $u = -1/2$ they are $(19/2, -19/2)$ and $(-19/2, 19/2)$ Wb. Thus physical sign changes and absolute sums vary with preparation while the differential trajectory is the same. A differential pickup that rejects c sees the same drive but retains its own backaction, load, and stores; its full coupled model is needed for the loaded comparison. Translating a reported flux origin alone changes none of the preparation works above.

The common-current preparation and pickup backaction therefore need their own source and load measurements [OP-LR42-02].

8.4 Commuting endpoints with noncommuting work partitions

Two capacitors C_1, C_2 exchange charge through one of two finite conducting channels A, B . The store boundary contains both capacitors; the active channel's heat reservoir is external. With $d = v_1 - v_2$,

$\bar{v} = (C_1 v_1 + C_2 v_2)/(C_1 + C_2)$, and $C_e = C_1 C_2/(C_1 + C_2)$, KCL gives

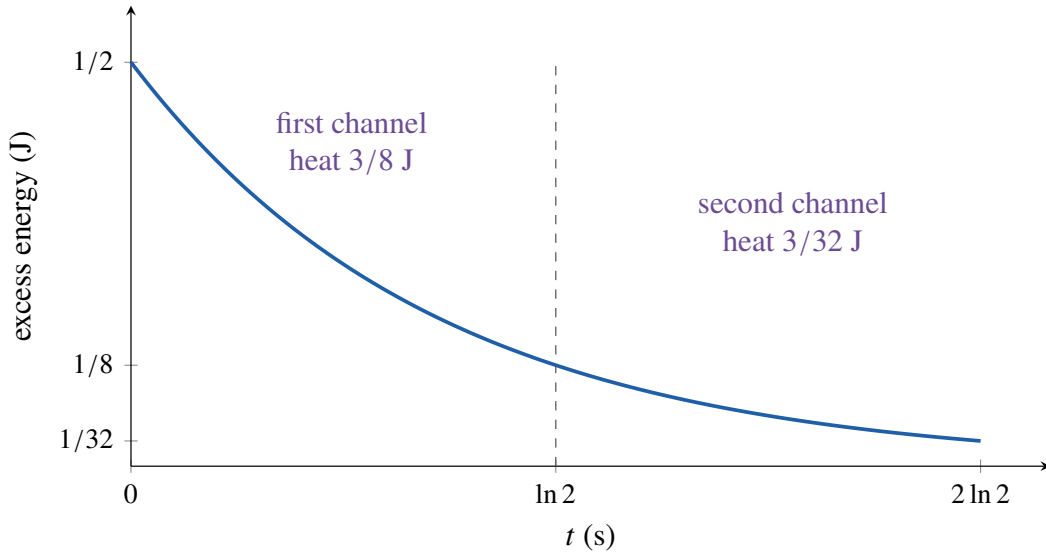
$$\dot{d} = -\frac{G}{C_e} d, \quad \dot{\bar{v}} = 0, \quad E = \frac{1}{2}(C_1 + C_2)\bar{v}^2 + \frac{1}{2}C_e d^2, \quad W_h = -\int_0^T G d^2 dt = -\frac{1}{2}C_e d_0^2(1 - e^{-2GT/C_e}). \quad (68)$$

Each stage is a finite $d \mapsto d/2$ map when $T = C_e \ln 2/G$. The two maps commute. Their heat integrals retain the incoming d and therefore the stage order.

For $C_1 = C_2 = 2$ F, $(v_1, v_2) = (1, 0)$ V, $G_A = G_B = 1$ S, each stage lasts $\ln 2$ s. Independently evaluated total stores are 1, $5/8$, and $17/32$ J before, between, and after the stages. Final voltages are $(5/8, 3/8)$ V in both orders.

Table 8. Equal endpoints do not determine the channel partition. Both controls have zero residual and no impulse at either finite switching edge.

Channel work into capacitor boundary, J	A then B	B then A
W_A	$-3/8$	$-3/32$
W_B	$-3/32$	$-3/8$
Total work	$-15/32$	$-15/32$
Final store	$17/32$	$17/32$



$A \rightarrow B$ or $B \rightarrow A$: same curve, different channel works

Figure 8. Exact excess-energy decay during two finite charge-sharing stages. The continuous state is the same for both orders, while the channel receiving the larger heat transfer changes. The plotted curve is equation (68).

Figure 8 separates the unchanged state curve from the changed heat recipient. Unequal-capacitance controls retain the common store. With $(C_1, C_2) = (1, 2)$ F, $C_e = 2/3$ F, the same voltage preparation and half-difference maps give channel works $-1/4, -1/16$ J and endpoints $1/2, 3/16$ J. With $(C_1, C_2) = (2, 1)$ F those works agree but endpoints are $1, 11/16$ J. Changing preparation to $(1, -1)$ V in the equal-capacitance case multiplies the excess store and both channel works by four, with common store zero.

Thus orientation and capacitance ratios must accompany a work claim.

Actual arc ignition, restrike, clamp release, winding capacitance, and switch supply laws are not determined by (68). A finite released winding/capacitor model can instead be written as $L\dot{i} = -v$, $C_p\dot{v} = i - \sum_l G_l(v, t)v$ with branch powers $-G_l(v, t)v^2$. Each proposed switch, arc, and clamp law is a separate G_l , with its own heat or receiver destination. Constant positive coefficients give exact matrix primitives; specified threshold switching uses event-side states and separate intervals. A constant-current or constant-voltage clamp uses its actual effort–flow law instead of this conductance representation.

For a connected perfectly coupled pair with $L = aa^T$, $a = (\sqrt{L_1}, \sqrt{L_2})$, and capacitor incidence $b = (1, -1)^T$, the equations $L\dot{i} = bv$, $C\dot{v} = -b^T i$ impose $n^T b v = 0$ for $n = (-\sqrt{L_2}, \sqrt{L_1})$. At $L_2/L_1 = 19$ this requires $v = 0$ and $i_1 = i_2$ on a smooth solution. The prepared state $(10, 0, 0)$ is incompatible; a putative terminal voltage impulse also has $n^T b \int v dt = 0$ and cannot select the null-current jump. Named parasitic paths can select finite releases, but different paths retain their different works. Other orientations require their own nullspace calculation.

Finite values $C_p = 1/100, 1/1000, 1/10000$ F and durations $1/50, 1/200$ s are separate realizable comparisons, conditional on their bandwidth and current envelopes. For explicit wider ratios, retain $(v_1, v_2) = (1, 0)$ V and both half-difference stages, but use $(C_1, C_2) = (1/2000, 1)$ F or $(1, 1/2000)$ F. The final stores are respectively $21/1334000$ and $10667/21344$ J; both export $5/21344$ J in total, from initial stores $1/4000$ and $1/2$ J. Each stage lasts $\ln 2/2001$ s at conductance 1 S. These finite ratios $1/2000$ and 2000 probe beyond the preceding range, conditional on their own measurement bounds; they do not certify a continuous limiting surface. At fixed prepared voltage, $C_p \rightarrow 0$ removes energy; at fixed nonzero charge, $E = q^2/(2C_p)$ diverges. At fixed R, C a prescribed voltage jump obeys (47); an RC charge-sharing path instead has finite limiting conversion but unbounded peak current as resistance vanishes. The exact zero-time and zero-component models remain mathematical questions separate from these finite tests.

Actual release, ignition, restrike and parasitic paths need independently identified laws before their channel partition is known [OP-LR48-03]. Finite ratios and finite edges test specified approaches, while other orientations and continuous limiting surfaces remain unproved [OP-LR48-05].

8.5 Different connections need not commute

Another scalar control has two independent 1 F cells initially at $(1, -1)$ V. Each selectable channel contains a separate 1Ω shunt across each cell, so its two laws are $\dot{v}_j = -v_j$. Operating a channel for $\log 2$ s halves both voltages. Two successive channels therefore end at $(1/4, -1/4)$ V and $1/16$ J from an initial 1 J. The independent two-resistor integrals give heats $3/4$ then $3/16$ J; reversing the channel order interchanges their recipients. These are shunts across individual cells; the earlier exchange resistor instead has time constant C_e/G . Stating the connections prevents a factor-of-two ambiguity in the duration of a half-difference map.

The scalar difference-mode maps above commute. With three stores a connection change can alter the final state as well as the recipient of heat. Take three 1 F cells with initial charges $q = (1, 0, -1)^T$ C. Channel A joins cells 1 and 2 through 1Ω ; channel B joins 2 and 3 . Each operates for $(\log 2)/2$ s. For incidence a , KCL is $\dot{q} = -aa^T q$ and its heat is $\int (a^T q)^2 dt$. The two exact maps are

$$M_A = \begin{pmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 3/4 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3/4 & 1/4 \\ 0 & 1/4 & 3/4 \end{pmatrix}. \quad (69)$$

Their ordered products and separately integrated channel heats give:

Table 9. Each final store is 133/256 J, evaluated from $q^T q/2$; each total heat is 123/256 J, from an initial 1 J.

Order	Final charges, C	Heat in A, J	Heat in B, J
A then B	$(3/4, -1/16, -11/16)$	3/16	75/256
B then A	$(11/16, 1/16, -3/4)$	75/256	3/16

Indeed $M_B M_A - M_A M_B \neq 0$. Equal energy here follows from the symmetric preparation; it does not restore equal final state. Finite resistor maps do not reset the cells exactly to zero. This bounded noncommuting extension leaves the physical partition under other connections and switch laws open [OP-LR48-02].

8.6 A damped electrical and mechanical preparation

A reciprocal port map extends beyond lossless oscillators. For a series circuit $\dot{q} = i$, $Li = V - q/C - Ri$ and a mass–spring–dashpot $\dot{x} = v$, $m\dot{v} = F - kx - bv$, set

$$q = \beta x, \quad i = \beta v, \quad V = F/\beta, \quad L = m/\beta^2, \quad 1/C = k/\beta^2, \quad R = b/\beta^2. \quad (70)$$

Then $Vi = Fv$, $Ri^2 = bv^2$, $q^2/(2C) = kx^2/2$ and $Li^2/2 = mv^2/2$. Substitution proves the state and port correspondence without assigning a balance residual as a missing work. Actual collision/contact laws and a manufactured realization remain unidentified [OP-LR50-01].

For $L = C = m = k = 1$, $R = b = 1/5$ and $\beta = 1$ in SI units, start at zero and apply effort +1 on $[0, 1]$, 0 on $[1, 2]$, then -1 on $[2, 3]$ s. The common state matrix is $A = \begin{pmatrix} 0 & 1 \\ -1 & -1/5 \end{pmatrix}$. On each interval the exact forced exponential determines its endpoint; the independent integrals $\int ui \, dt$ and $-\int i^2/5 \, dt$ are given by (2) after adjoining a constant state. The two physical models therefore have identical signed works and stores on each finite interval. The third endpoint is retained; opposite drive is not an assumed exact reset.

A physical common-motion preparation also has a source. Over $t \in [-1, 0]$ s, put $s = t + 1$ and prescribe $x = (3/10)(s^3 - s^2)$ m, $v = \dot{x}$, $F = \ddot{x} + \dot{x}/5 + x$ in the same unit model. Integration of the force and dashpot products gives

$$W_F = \frac{237}{5000} \text{ J}, \quad W_b = -\frac{3}{1250} \text{ J}, \quad E(-1) = 0, \quad E(0) = \frac{9}{200} \text{ J}. \quad (71)$$

The endpoints $(x, v) = (0, 0)$, $(0, 3/10)$ are evaluated independently. Changing an observer coordinate alone performs none of this preparation.

9 Material, thermal, sensor, and supply boundaries

9.1 Conversion is different from heat already exported

For a fixed physical mesh, opposing sliding forces convert $D = \mu N |v_{\text{slip}}|$. Both contact points receive the same observation velocity subtraction, so v_{slip} and D are invariant. For $\mu = 1/2$, $N = 2$ N, $|v_{\text{slip}}| = 1$ m/s over one second, the mechanical conversion is 1 J in every observation; at $\mu = 0$ it is zero. Bearings use relative angular speed, for example $D_b = b(\omega - \omega_b)^2$, while a declared quadratic drag effort gives windage conversion $D_w = c|\omega - \omega_a|^3$. Lubricant churning needs its own measured constitutive law and moving-fluid boundary. None follows from the ideal rolling equation.

A thermal body with heat capacity C_T , excess temperature θ above a fixed ambient $T_a > 0$, and

conductance H_T obeys

$$C_T \dot{\theta} = D - H_T \theta, \quad E_T = C_T \theta, \quad P_{\text{amb}} = -H_T \theta = -T_b \dot{S}_{\text{out}}. \quad (72)$$

For constant D and initial $\theta = 0$, $\theta = (D/H_T)(1 - e^{-H_T t/C_T})$. Choose $D = 1$ W, $C_T = H_T = 2$ in SI units on one second. The independently integrated internal conversion, external heat, and store change are $+1$, $-e^{-1}$, and $1 - e^{-1}$ J. Immediate export would predict -1 J and zero thermal change. In the combined mechanical/thermal boundary the conversion cancels between the two subsystems; only ambient heat crosses the combined boundary. For repeated identical heating intervals the exact map is $\theta_{n+1} = e^{-H_T T/C_T} \theta_n + (D/H_T)(1 - e^{-H_T T/C_T})$. No equality of initial and final temperatures is assumed. If copper resistance varies with temperature, specify $R(T)$ independently and integrate $R(T)i^2$ on the actual path; a fixed-temperature law cannot identify that feedback.

Coulomb conversion is invariant under a change of observation for the same physical contact; the actual friction law still requires identification [OP-EPI-13]. Bearing drag, windage, churning and retained heat require their own ports and endpoint observations [OP-EPI-25].

9.2 Magnetic material states and finite settling

A normalized but dimensionally scaled magnetic mode in configuration VI has flux ϕ , material state m , and energy

$$E_m = \frac{\phi^2}{2} + \frac{a\phi^4}{4} + \frac{h(\phi - m)^2}{2}, \quad i = \frac{\partial E_m}{\partial \phi} = \phi + a\phi^3 + h(\phi - m), \quad \dot{\phi} = v - Ri, \quad \dot{m} = \frac{h(\phi - m)}{\zeta}. \quad (73)$$

Positive a, h, ζ carry the units making each term an energy or the stated electrical variable; numerical controls set their SI coefficients to one where specified. Differentiation from these independent laws gives $\dot{E}_m = vi - Ri^2 - h^2(\phi - m)^2/\zeta$. Thus electrical input, copper heat, and material heat are distinct integrals. Positive a gives a saturation-type increasing reluctance; a single lag is one admissible rate-dependent model, not an identified magnetic material.

For a reversible ramp, $\phi = t$, $h = 0$, $R = 1/5$ on $[0, 1]$ s, $i = t + at^3$, $v = 1 + Ri$, and initial energy zero. With $a = 0$ the separate source and copper works are $17/30$ and $-1/15$ J, giving final energy $1/2$ J. With $a = 1$ they are $3/4 + 92/525$ and $-92/525$ J, giving $3/4$ J. The extra magnetic input is $1/4$ J, calculated from $\int i \dot{\phi} dt$ for each path. This does not specify hysteretic loss.

For an exact periodic lag control take $a = h = \zeta = 1$, $R = 1/5$, $\phi = \sin t$, $m = (\sin t - \cos t)/2$ and impose $v = \cos t + Ri$ on $[0, 2\pi]$ s. Direct integration gives

$$W_s = 63\pi/40 \text{ J}, \quad W_{\text{copper}} = -43\pi/40 \text{ J}, \quad W_{\text{material}} = -\pi/2 \text{ J}, \quad E_m(0) = E_m(2\pi) = 1/8 \text{ J}. \quad (74)$$

For the reversible $h = 0$ control on the same imposed ϕ , magnetic cycle input and material heat are both zero; its source still supplies its own copper integral. The lag control's initial $m = -1/2$ is a prepared periodic state. Its preparation work is not assigned from $1/8$ J.

Finite settling can be checked exactly. Set $m = (\sin t - \cos t)/2 + \delta e^{-t}$ on the same interval. Material conversion becomes $\pi/2 - \delta(1 - e^{-2\pi}) + \delta^2(1 - e^{-4\pi})/2$; magnetic electrical input is $\pi/2 - \delta(1 - e^{-2\pi})/2$. The endpoint stores, evaluated from the material state, are $(1/2 - \delta)^2/2$ and $(1/2 - \delta e^{-2\pi})^2/2$ J. Their change is $\delta(1 - e^{-2\pi})/2 - \delta^2(1 - e^{-4\pi})/2$ J. Measure $\int i(v - Ri)dt$, flux linkage, thermal transfer, and endpoint material indicators over successively specified finite periods. Saturation, remanence and hysteresis laws require independent identification; a loop that has not returned to its initial material state

is not an exactly periodic heat measurement.

Finite magnetic constitutive behavior must be identified beyond the linear winding law [OP-TRF-03]. Remanence, hysteresis and temperature-dependent states in an actual material retain a separate cycle-work question [OP-LR57-01].

9.3 Finite sensors and biased electromechanical actuation

A sensor capacitor driven by a held unit voltage through $R = 1 \Omega$, $C = 1 \text{ F}$ from zero has $v = 1 - e^{-t}$ and source current e^{-t} . On $[0, \ln 2]$ s, the source work is $1/2 \text{ J}$, resistor heat work $-3/8 \text{ J}$, and capacitor end-point store $1/8 \text{ J}$. Their separate products are $1 \cdot e^{-t}$ and $-e^{-2t}$. A nonloading ideal observation assigns all three zero; this finite sensor has both delay and supplied energy. If its resistor is inside a thermal boundary, use (72) instead of exporting its conversion twice. Real probe input capacitance, transducer bias, electronics supply, and calibrated transfer function belong to the sensor model. Noise, delayed or nonlinear observers, and multiple feedback channels are additional states or error processes, not consequences of this scalar control.

For the electromechanical plant, independently specify

$$L\dot{i} = v_s - Ri - g_e v_m, \quad \dot{x} = v_m, \quad m\dot{v}_m = g_m i - c v_m - kx + F_s. \quad (75)$$

Its store is $Li^2/2 + mv_m^2/2 + kx^2/2$. Multiplication by i and v_m identifies the separate controller transfer

$$P_a = (g_m - g_e)iv_m, \quad \dot{E}_{\text{plant}} = v_s i + F_s v_m - Ri^2 - cv_m^2 + P_a. \quad (76)$$

The controller receives $-P_a$ on the opposite boundary. Reciprocal coupling has $g_m = g_e$; unequal or reversed backaction requires a separately supplied physical realization, not a negative store.

For an exact finite supply control use $L = m = 1$, $R = c = k = 0$, $g_e = 1$, $g_m = 2$, $i = v_m = 1$, $v_s = 1$, $F_s = -2$ on one second. The imposed motion satisfies (75), its store is 1 J at both endpoints, and separate works are $W_e = 1$, $W_m = -2$, $W_a = 1 \text{ J}$. Let a declared bidirectional converter have efficiency $1/2$ in delivery, so that delivering $P_a = 1 \text{ W}$ draws 2 W and converts 1 W to heat. A bias coil $L_b = 1/2 \text{ H}$, $R_b = 1 \Omega$, held at $i_b = 1 \text{ A}$ requires another 1 W and has store $1/4 \text{ J}$ at both endpoints. The source-to-controller work is 3 J ; output, converter heat, and bias heat works are each -1 J . The controller therefore has zero store change by its independent states.

Realize the finite source by a rail $C_s = 2 \text{ F}$, $V_s(t) = \sqrt{4 - 3t} \text{ V}$, with ideal regulated draw $I_s = 3/V_s \text{ A}$. Its store changes from 4 to 1 J , and its separately integrated output is $-\int V_s I_s dt = -3 \text{ J}$. The combined plant, controller, and rail endpoints are $21/4, 9/4 \text{ J}$; external electrical input, receiver output, and the two heat works sum to $1 - 2 - 1 - 1 = -3 \text{ J}$. A remaining voltage above the declared cutoff 1 V permits this interval; beyond its endpoint the regulator law must change. The regulated draw and conversion law are assumptions requiring their own actuator and supply measurements.

Source-off does not imply passive decay of every controller store. For a separate return control, a mechanical actuator delivers 1 W to a converter with recovery efficiency $1/2$ for one second. The rail receives $1/2 \text{ J}$, converter heat receives $1/2 \text{ J}$, and the actuator receives work -1 J . If $C_s = 2 \text{ F}$ and $V_s(0) = 1 \text{ V}$, then $V_s(t) = \sqrt{1 + t/2}$ and rail current is $1/(2V_s)$; direct integration gives the stated positive rail work. This is an explicitly supplied return path, not a prediction for an unidentified bias coil. The actual bias law can make coupling coefficients depend on i_b and must be solved with the plant. A bounded algebraic controller or a positive aggregate store cannot certify its separate coil, rail, thermal, or whole-apparatus work accuracy.

Finite sensors and thermal endpoints require measured loading, supply work and retained states [OP-LR57-07]. The realized actuator and bias supply must support each forward or returned controller transfer [OP-LR59-02].

9.4 Complete cycles and what a prescribed path establishes

For every actual apparatus keep stages ℓ distinct: preparation, operation, switching, relaxation, reset, and any hold interval. The measured statement is

$$r_{\text{cycle}} = E_{\text{final}} - E_{\text{initial}} - \sum_{\ell} \sum_j \int_{t_{\ell,0}}^{t_{\ell,1}} e_j f_j dt. \quad (77)$$

A periodic thermal state, restored battery state, or emptied capacitor is an endpoint observation, not a premise. For a source-free regular linear relaxation, $x(T) = e^{AT}x(0)$ and the exponential is invertible. A nonzero state cannot become exactly zero at finite T by that law alone. For a capacitor C , V_0 relaxed through R for $T = RC \ln 2$, separate heat work is $-3CV_0^2/8$ and endpoint store $CV_0^2/8$.

A completely specified elementary 2 J receiver cycle uses a unit capacitor. On preparation $[0, 2]$ s impose $v = t$ with source current 1 A; its work is +2 J and store rises from 0 to 2 J. On operation $[2, 6]$ s use an ideal controlled receiver with $v = (6 - t)/2$, receiving current 1/2 A. Its work into the capacitor boundary is -2 J and store falls to zero. On reset $[6, 10]$ and hold $[10, 12]$ s, $v = i = 0$ and all works are zero. Currents change finitely at the switches; capacitor voltage is continuous. This is a prescribed, lossless current-control model. A one-half-efficient physical preparation converter instead draws 4 J to deliver the same 2 J preparation; its independent heat work is -2 J. Both predict the same receiver transfer and different measured source work.

A finite reset compares further paths. Discharging a prepared unit-energy store to zero through a resistor sends 1 J to heat in a complete asymptotic reset, with finite endpoints retained on every finite interval. A lossless resonant auxiliary receives that energy in a finite quarter-period and must return it during a specified later preparation to close its own cycle. A one-half-efficient regenerative converter delivering the same 1 J removal to a battery stores 1/2 J and converts 1/2 J to heat; drawing that 1/2 J back through the same efficiency returns only 1/4 J to the plant. Battery terminal work, chemical state, and temperature must be independently specified; those efficiencies are illustrative laws, not battery measurements.

For the broader physical cycle questions, retain the different apparatus. A differential three-coil pickup, rectification before or after combining pickup signals, and a tuned receiver each have separate preparation and reset connections. Finite paths for those systems do not establish cycles for a local/distant-field winding arrangement, a moving or relaxing magnetic material, or a spark-gap resonator. For the field arrangement integrate every winding source and reset branch and include field energy endpoints. For motion/material apparatus add force-velocity and material/thermal states. For a spark-gap resonator retain primary, secondary, gap, clamp, and parasitic capacitor ports on each event side. A 2 J receiver target can be held for each comparison, but its required source work is $2J + Q_{\text{ext}} + \Delta E - W_{\text{other}}$, where each term is independently observed or predicted from its specified law; this identity is not a way to assign an unmeasured port. No unspecified source, arc, battery, or controller law has a definite efficiency or cycle work here.

A complete transformer cycle requires all finite stages and observed reset states [OP-TRF-05]. Prescribing a path does not establish a physical supply's work or feasibility [OP-LR49-01]. The broader field, pickup, moving-material and spark-gap apparatus retain their own missing laws [OP-LR49-02].

10 What must be measured

10.1 Independent uncertainty for work and endpoints

Record conjugate effort and flow on the same physical interval, including all event sides. If true channels differ from recorded $\hat{e}, \hat{f}$ by at most u_e, u_f , an exact bound is

$$|W - \widehat{W}| \leq \int_{t_0}^{t_1} (|\hat{e}|u_f + |\hat{f}|u_e + u_e u_f) dt + U_{\text{time}} + U_{\text{band}} + U_{\text{read}}. \quad (78)$$

Here the additional bounds cover integration endpoints and relative timing, unresolved signal bandwidth, and reading reconstruction, including noise and quantization. They are established independently of a small energy residual. For a finite endpoint error δt and bound $|P| \leq P_{\text{max}}$, the omitted endpoint work is at most $P_{\text{max}}|\delta t|$. If a flow channel has $|\dot{f}| \leq B_f$, a relative timing offset bounded by u_t contributes flow error at most $B_f u_t$ by the mean-value theorem. Neither bound applies across an unresolved ideal impulse; use the finite physical pulse and its actual envelopes.

For a capacitor with recorded positive $\widehat{C}$, $|C - \widehat{C}| \leq u_C$, and $|V - \widehat{V}| \leq u_V$, the exact endpoint bound is

$$U_E \leq \frac{u_C}{2} (|\widehat{V}| + u_V)^2 + \widehat{C} \left(|\widehat{V}|u_V + \frac{u_V^2}{2} \right). \quad (79)$$

Use the quadratic-form analogue for coupled magnetic or mechanical stores, including covariance or interval dependence of their parameters and states. The mutual term is not two independent inductors. Thermal endpoints require heat capacity, temperature distribution, and heat-transfer uncertainties; material endpoints require the stated internal-state law. A residual bound is

$$U_r = U_{E_0} + U_{E_1} + \sum_j U_{W_j}, \quad r_E \in [\hat{r}_E - U_r, \hat{r}_E + U_r]. \quad (80)$$

This conservative interval needs no independence assumption. Correlation can sharpen it only when established by calibration and propagated jointly. For probabilistic intervals, specify the joint error law and coverage; for certified intervals, enclose its support and nonlinear products. General metrological treatment of inputs and correlations is described by [JCGM \(2008\)](#); the finite product and endpoint bounds here are derived directly and discard no second-order term.

Phase and polarity errors illustrate the size of the problem. For peak signals $v = \cos t$, $i = \cos(t + \phi)$ over $[0, 2\pi]$ s,

$$W = \pi \cos \phi \text{ J}, \quad \widehat{W} = \sigma(1 + a)(1 + b)\pi \cos(\phi + \delta) \text{ J}. \quad (81)$$

Here $\sigma = \pm 1$ is channel polarity, a, b are gains, and δ is relative phase including timing at the stated angular frequency. At $\phi = \pi/2$, $\delta = \pi/6$, unit gains and correct polarity, $W = 0$, $\widehat{W} = -\pi/2$ J. At $\phi = 0$, $a = 3/100$, $b = -3/100$, $\delta = 0$, the gain error is $-9\pi/10000$ J. Reversing only current changes $+\pi$ to $-\pi$ J. Reversing both channels does not do so. These predictions follow by direct trigonometric integration. A sensor voltage error 1/100 V at a true 100 V on a 1 F capacitor gives endpoint error 20001/20000 J. Similar-looking endpoint voltages do not establish cancellation of their calibration errors.

A low-pass channel's transfer, loading, time alignment, noise spectrum, and quantization must be calibrated over the signals being integrated. In a multi-port assembly, moving a probe across a shunt can include a different branch: the extra current is a physical port-placement error with correlated voltage and store consequences. Synchronizing instruments alone does not bound it. A sensor that loads the apparatus

changes its trajectory and belongs in its constitutive comparison.

The calibrated polarity, gain, phase, timing, bandwidth and endpoint uncertainties determine the signed residual that can be resolved [OP-LR63-02].

10.2 Finite comparisons and decision requirements

In the following tables, every pair gives different predictions for at least one accessible quantity under its stated control. A zero in the first column may be a null hypothesis about an omitted contribution, not an established physical law. An exact alternative applies only with its explicitly stated constitutive assumptions. For a pair separated by $\Delta > 0$, require a combined prediction-setting and observation uncertainty $U < \Delta/2$; $U \leq \Delta/4$ is a concrete design target. This requirement is applied to the individual comparison, never inferred from instrument names. If the required bound is unavailable, that comparison remains unresolved. A finite outcome can reject one of these readings on its interval; it does not close the same question for every material, geometry, or drive.

For an unknown physical remainder, the complete-model reading is $r_E = 0$; a surviving-discrepancy reading is $r_E = \delta$ with $|\delta| > U_r$, where sign and magnitude must be measured. These are distinct boundary predictions but there is no justified fixed nonzero value before measurement. To validate the observation chain at a known scale, introduce an independently metered 1/4 J positive input. Omitting it predicts $r_E = +1/4$ J; including it predicts zero for the otherwise exact control. Requiring $U_r < 1/8$ J distinguishes that omission. This calibration control is not a prediction of a physical energy anomaly.

The loaded carrier comparison has a practical conditional budget. Suppose both torque records have $|\widehat{\tau}| \leq 8$ N m and $|\widehat{\omega}| \leq 4$ rad/s, with respective pointwise errors 1/100 N m and 1/100 rad/s. On two one-second observations the product part of (78) is at most 1201/5000 J. A total comparison bound of 1 J leaves 3799/5000 J for the other contributions and separates the 7 J attachment difference. It does not establish a 1 J full-boundary residual uncertainty.

For the reference actuator, $\widehat{C} = 1$ F, $u_C = 1/100$ F, $|\widehat{w}| \leq 2$ V, and $u_V = 1/100$ V give $U_E \leq 80501/2000000$ J at each endpoint. Two final-state observations have total 80501/1000000 J before setting and timing uncertainty. A 1/10 J comparison budget leaves 19499/1000000 J for those effects. It resolves $[(9 + e^{-10})^2 - 100]/50$ J from zero: since $e^{-10} < 1/100$, its magnitude exceeds 188199/500000 J. For full work closure also observe actuator, sensor, driver, rail, and thermal endpoints or their external supply products.

10.3 Mechanical comparisons

Hold the declared rates or paths with separately metered actuators and observe torque and angle, force and displacement, and each endpoint motion together. The table preserves distinct contact, support, and whole-system boundaries.

Table 10. Distinct mechanical measurement targets. The deliberate extra-torque and extra-work alternatives are explicit falsification controls, not assigned properties of unknown joints or friction laws.

Question and held control	Reading A	Reading B and decisive observation
Stator mounting, same negative-speed motion	No change in conversion: 0	$Q_c - Q_0 = 7\pi/50$ J; integrate drag torque against each relative angle, retaining phase endpoints
Attached receiver, configuration I	Unloaded carrier work −7 J	Loaded 0 J; measure sun, ring, carrier, and all three receiver works on the same motion

Question and held control	Reading A	Reading B and decisive observation
Transient frame closure	Carrier shafts alone sum to 595/72 J	Effective change 517/72 J with field work $-13/12$ J; reconstruct all four observations from synchronized motions and inertias
Location of frame difference	Held-ring work remains 0 in both observations	Carrier-ring work -5 J in steady operation; integrate its independently measured reaction torque
History partition, same endpoints	Shaft difference 13/12 J at $b = 0$	13/9 J at $b = 1$; drive the distinct intermediate ring path and observe both field and shaft shares
Hooke support over a complete cycle	Torque-path $W_u = 0$	A specified extra unit support torque adds $\pi/2$ J over the control cycle; resolve its torque and support angle rather than assign it from the residual
Ratio-two bevel support	Held support flow gives 0	Moving support gives $-\pi/2$ J in ground; carrier reconstruction gives zero on the same cycle
Compound observation, close-ratio geometry	Ground $\mathcal{C}_0 = 1$	Carrier 2950/21; measure each signed shaft and contact side and bound the nonzero ground denominator
Other geometry, two-sun control	Ground shaft ratio 1	Carrier shaft ratio 6/5; instrument four shafts and retain all four meshes; helical or bevel thrust requires its own force/motion observations
Compliant engagement	An inferred instantaneous 1 J jump from a power change	Actual instantaneous store jump 0 despite a 1 W power jump; resolve finite contact work 2 J, heat 1 J, and spring endpoint 1 J
Restitution control	Elastic $e = 1$: kinetic loss 0	$e = 1/2$: loss 3/16 J; observe incoming/outgoing motion and resolved contact work, with adequate pulse bandwidth
Compliance with acceleration	Omitting the field leaves carrier final accounting at 27/32 J	Including $-1/2$ J field work gives 11/32 J effective store in the six-contact train; observe all contact states and event sides
Unequal planets and pin couple	Equal first-contact work 7/3 J; zero pin-pair work	First contact 7/2 J; the separate one-pin control gives $-1/3$ J; measure individual forces and pin torques
Locked approach	Drive-scaled limiting works (0, 0, 0)	Fixed-drive (2, 5, -7) J; record drive and relative motion separately at each finite approach
Mesh friction invariance	Frame heat difference 0	A postulated frame-dependent extra 1/4 J differs from zero; compare reconstructed relative-slip products on the same contact, not remounted friction bodies

Question and held control	Reading A	Reading B and decisive observation
Bearing/windage/churning with a thermal store	Immediate export 1 J, retained heat 0	Unit thermal control exports e^{-1} J and retains $1 - e^{-1}$ J; observe temperatures and ambient heat as well as mechanical conversion
Radial motion	Omitted radial work 0	Radial guide work $-3/2$ J; measure radial force and displacement with tangential support work $+3$ J
Nonparallel observation	Scalar-only field work 0	Vector control -1 J; reconstruct all angular-momentum components and observer motion
Within-interval reversal	Net signed work 0 taken as zero traffic	Absolute transfer $1/2$ J and unit-drag heat $1/2$ J; integrate both sides of the flow zero
Cardan transverse and bending work	Fixed application point: 0	Unit force through unit displacement: 1 J; determine actual bearing constraints and all force/couple conjugate motions
Unequal angles, acceleration, phase actuation	Equal-angle partial support, steady rotor pin, and absent phase drive: (0, 0, 0)	Separate controls give $(\pi/4 - \arctan(3/4), 1/2, 1)$ J; observe each support and phase supply on its own interval
Physical gear closure	Complete physical residual 0	δ outside its independent interval bound; validate the chain with the $1/4$ J input, then retain any actual signed discrepancy with all completed checks

A physical gear remainder has unknown magnitude and sign until the complete boundary is measured within those independent uncertainties [OP-EPI-34].

10.4 Electrical and correspondence comparisons

Use synchronized complete voltage/current pairs for source, receiver, probe, reference driver, switch, and heat-conversion channels. Independently observe winding currents, capacitor voltages, flux or identified material states, and thermal endpoints. A specified finite waveform must fit the calibrated channel bandwidth without clipping; opening pulses require a separate bound.

Table 11. Electrical comparisons. Finite-model success includes every port and both state endpoints on each interval; neither an ideal ratio nor a small aggregate remainder supplies a missing dynamic map.

Question and control	Reading A	Reading B and decisive observation
Physical reference attachment	Common coordinate shift: added work 0	Unit RC attachment requires $5/6$ J input, retains $1/2$ J; observe its actual return current and voltage
Complete transformer cycle	Passive finite relaxation called reset: final store 0	At $RC \ln 2$, final store $CV_0^2/8$; observe preparation, receiver, switch, supplies, and all remaining states separately

Question and control	Reading A	Reading B and decisive observation
Nonlinear magnetic/thermal state	Linear unit ramp: magnetic input $1/2$ J	Saturated ramp: $3/4$ J; observe $i(v - Ri)$ and actual material/thermal endpoints
Energized winding opening and return commutation	Unit-voltage clamp receives $3/2$ J	Two-volt clamp receives 2 J; observe each clamp and switch port; the tapped overlap/gap alternatives use their separately defined matrix works
Finite reference realization	Ideal receiver endpoint 2 J	Finite actuator $(9 + e^{-10})^2/50$ J; compare source products, tracking, actuator stores, and finite rail state
Constrained winding limits	Zero-store ideal transformer at the unit ramp: 0	Perfect-coupling finite-magnetizing model: $1/2$ J; retain the separate capacitance, source-resistance, short, and zero-ratio controls in the constrained table
Crossed load, probe, frequency, and preparation	Prepared RC receiver gets $(1 - e^{-2})/2$ J without the probe	With probe it gets $(1 - e^{-4})/4$ J; use all 16 specified winding settings for interaction claims and each exact work primitive on its own state
Physical transformer closure	Complete physical residual 0	δ outside its independently bounded interval; the known $1/4$ J input validates detection separately from physical closure
Loaded dynamic mapping	Coordinate-shift-only capacitor change 0	Supplied compensation gives 2 J, matching the relative kinetic control; observe the extra ports and state mapping, not solely terminal counts
Ideal compound terminal versus physical ratio	Complete-port ratio 1 for $R = -341/100$, B held	A-frame mapped ratio $341/100$; observe actual winding voltages and reconstructed terminal contributions separately

The corresponding physical transformer remainder requires simultaneous port and endpoint observations, including switch, probe and supply effects [OP-TRF-07].

10.5 Preparation, switching, material, supply, and calibration comparisons

These comparisons retain the original apparatus classes. A general work law can be shared; a measured outcome for one device cannot be transferred to a different winding, field, impact, pickup, or controller arrangement.

Table 12. Prepared and supplied controls. The source-off rail comparison, hidden-mode control, and material loop belong to their explicitly stated systems; they do not certify wider apparatus or unmeasured constitutive laws.

Question and control	Reading A	Reading B and decisive observation
Prepared singular-component paths	Zero-state two-section circuit: heat 0	Prepared state in equation (63): heat $(1 - e^{-2})/2$ J; cross finite component scalings and observe both independent states
Switching work beyond smooth zero-state transfer	Resistive regularization exports $3/8$ J by its declared endpoint	Reactive quarter-period transfer exports zero heat and leaves $1/2$ J in the auxiliary; integrate all ports on each finite path
Nearly cancelled, multiple, or nonideal modes	One known mode: at most $1/20000000000$ J under the stated current bound	Two modes at ± 1 V: $1/1000000$ J despite zero terminal signal; resolve internal branches, coupling uncertainty, and observation rank
Exact hidden mode	Empty branch: initial energy and release heat 0	Prepared microfarad branch: $1/2000000$ J and $3/8000000$ J; observe branch current/voltage and retained endpoint
Physical common-mode preparation	$u = -1/2$: input $95/2$ J and no half-period coil exchange	$u = 1/2$: input $115/2$ J and coil works $(-19, +19)$ J; observe preparation sources and flux signs, retaining differential load backaction
Same switching endpoint, different partition	A then B: heat pair $(3/8, 3/32)$ J	B then A: $(3/32, 3/8)$ J; measure channels separately, not only total heat or final voltages
Actual switch/gap and release laws	Unit clamp: $3/2$ J receiver transfer	Two-volt clamp: 2 J; distinguish ignition, restrike, overlap, gap, and release states through their branch products and endpoint capacitances
Finite capacitance ratios/orientations and limiting surfaces	$(C_1, C_2) = (1, 2)$ F: final store $3/16$ J	Reversed ratio $(2, 1)$ F: $11/16$ J; extend to declared finite smaller parasitics and wider ratios with their actual bandwidth; no finite outcome realizes zero time
Controlled cycle versus physical supply	Lossless 2 J preparation requires 2 J source input	Half-efficient preparation requires 4 J; observe supplies, receiver, conversion and thermal endpoints on the complete schedule
Missing preparation/reset and converter laws in other apparatus	Ideal recovery of a 1 J removal stores 1 J in the receiver	Half-efficient recovery stores $1/2$ J; observe every actual source, field/mechanical, gap, battery, and reset path for that apparatus
Identified hysteretic material and settling	Reversible magnetic cycle input 0	Periodic lag control $\pi/2$ J; measure internal-state closure and copper-subtracted magnetic input over declared periods

Question and control	Reading A	Reading B and decisive observation
Realized thermal and sensor ports	Nonloading ideal sensor input and stored energy 0	Finite sensor input $1/2$ J and store $1/8$ J; record sensor supply, loading, temperature, and delay before extending to other feedback structures
Finite electromechanical bias supply	Treating unequal backaction as supplied at zero work misses 1 J	Plant receives 1 J actuator work, controller draws 3 J including its declared losses; source-off recovery control raises rail energy $1/2$ J
Channel-specific uncertainty	True quadrature work 0	Relative phase error gives $-\pi/2$ J; inject known polarity/gain/timing changes and calibrate joint port-placement and endpoint errors separately

A comparison closes its bounded question only when the actual domain, connections, event law, source settings, endpoint states, and uncertainty meet the stated conditions. A measured mismatch is retained with sign, magnitude, interval, and completed polarity, loading, timing, endpoint, and model checks. Missing constitutive identification or inadequate resolution leaves the question open. A finite family of successful comparisons does not establish universal absence of a discrepancy.

11 Relation to earlier work and limits of the mathematics

The carrier-relative gear construction in [Culpepper \(2002\)](#) and the rotating-coordinate mechanics in [MIT \(2022\)](#) support the starting kinematics. The individual work differences here follow from explicit free bodies and exact integrals. They do not turn a rotation count into a work measurement. The magnetic terminal relations in [Haus and Melcher \(1989\)](#) distinguish finite inductive storage from an ideal algebraic winding relation. The physical return, clamp, and supply comparisons require further circuit laws given here. The metrological framework of [JCGM \(2008\)](#) motivates recording input quantities and correlations; it does not provide a numerical uncertainty for this apparatus.

The common ground with [Nilre and Herlin \(2026\)](#) is the separation of complete physical ports, observation-dependent contributions, and independent state endpoints. The additional controls expose contact, topology, preparation, hidden-state, material, and calibration assumptions needed for finite comparisons. Every theorem above is conditional on its equations. The simple rolling and axial linkage results do not settle actual Cardan transverse reactions. The Ravigneaux pitch example does not settle helical thrust or every nonparallel contact. Finite compliant and restitution controls do not identify a real impact law. Ideal terminal correspondence and the elastic dynamic map have different state spaces and different closure domains.

Exact polynomial or trigonometric controls can locate every power zero algebraically. A root-free interval, a tangent zero, a persistent identity, and equality of maximum absolute powers have different roles. A complete event description for one specified smooth constitutive model does not supply nonlinear switch laws for another. A same-sign narrow transfer may also carry substantial work without any interior power zero. The power integrals and their bandwidth requirement remain independent of event counts.

Physical magnetic identification, finite-source realization, and full cycle closure remain empirical questions. Bounded results for prepared paths and smooth-model correspondences leave their wider do-

mains intact. The numerical integration residual here is exactly zero, and each complete ideal control has $r_E = 0$ by independently calculated works and endpoints. Physical model residuals remain unmeasured. They include omitted joint, contact, bearing, windage, churning, switch, arc, magnetic, dielectric, thermal, sensor, controller, and supply behavior. No aggregate model identity assigns the magnitude or destination of an unexplained transfer.

12 Conclusion

The fourth shaft turns a readout question into a loaded-machine test. In the ideal one-receiver control, the output takes $7/3$ J and carrier work changes by the same amount while the held ring carries torque without ground work. The free bodies determine that result. Actual U-joints require measured spatial reactions and application-point motions. A dimensioned Oldham model, a finite clutch and specified stator loads establish their own transfers and show which assumptions change.

The switched electrical apparatus has equally specific connections. Full-to-tap return changes the circuit; its winding, cell, gate and thermal states persist through the event. A clocked contraction theorem leaves hybrid guards and material admissibility to their own analysis. Generator comparisons require prime-mover, field and controller work over matched service with actual preparation and reset.

Two series banks can share an entire terminal history while retaining 1 or 5 J. Reconnection can distinguish those known states under a specified observation map, but receiver delivery follows its separate integral and need not track the hidden-energy ordering. The LC/shunt certificate establishes every charge-sign event for its declared window. Three-store reset changes the final state with order; the damped correspondence gives a physical preparation whose source work is independently calculated. These are bounded results with explicit state and topology domains.

The remaining experiments require measured laws, resolved work differences and independent endpoint uncertainties. A small algebraic residual supplies none of those observations. Any positive or negative physical remainder retains its measured value, conditions and uncertainty until its cause is identified.

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